

RIX2024 - Batch 1 – Could Computing Practical Assessment
12576 Nisanth Saravanan

Peer-to-Peer Learning Exchange

Abstract: Users exchange skills through scheduled learning sessions.

Entities: User, Skill, Session

W3H Analysis

Project Title: Peer-to-Peer Learning Exchange	
What?	How?
1. What are the necessary modules? A: Modules Needed: 1) Login and Registration. 2) User Dashboard. 2. What are the necessary functionalities of Login and Registration Modules? A: Login and Registration Module: 1) Register User. 2) Login as User. 3. What are the necessary functionalities of User Dashboard? A: User Dashboard: 1) Create, Update, Edit and Delete Sessions. 2) Create, Update, Edit and Delete Skills. 4. What are the tables needed? A: Tables Required: • Users • Skills • Sessions	Login and Registration Module: 1) Register User: Method 1: Use Simple Form Page. Method 2: Use a Modular Form Page. Method 3: Use Google OAuth and other 3rd party services. 2) Login as User: Method 1: Use Simple Form. Method 2: Use a Modular Form. Method 3: Use 3rd party services. User Dashboard: 1) Create, Update, Edit and Delete Sessions: Method 1: Use Simple Forms and Buttons to perform CRUD. Method 2: Have separate pages for each operation Method 3: Use Modals. 2) Create, Update, Edit and Delete Skills: Method 1: Use Simple Forms and Buttons to perform CRUD.

5. What are the fields of Users table? A: Fields of Users table: • Id (Primary Key) - int • Name – string • Contact – int 6. What are the fields of the Skills table? A: Fields of Skills Table: • Id (Primary Key) - int • SkillName – string • Proficiency - int 7. What are the fields of the Session table? A: Fields of Session Table: • Id (Primary Key) - int • Skill – string • DateTime – DateTime 8. What are the validations in the application? A: Overall validations list: • Frontend simple JS Validations. • Server Side validations. • Form Validations.	Method 2: Have separate pages for each operation Method 3: Use Modals.
Login and Registration Module: 1) Register User: Method 1: Use Simple Form Page. A: Since the overall scale of the application is small and data needed is rather minimal, it's wise to stick with traditional forms. 2) Login as User: Method 1: Use Simple Form. A: Since the overall scale of the application is small and data needed is rather minimal, it's wise to stick with traditional forms. User Dashboard: 1) Create, Update, Edit and Delete Sessions: Method 1: Use Simple Forms and Buttons to perform CRUD. A: Since the overall scale of the application is	Login and Registration Module: 1) Register User: Method 2: Use a Modular Form Page. A: Increases the complexity of the app, and also the fact that the number of needed inputs in rather minimal. Method 3: Use Google OAuth and other 3rd party services. A: Valid method but due to the scale of the application, this method is seemingly overkill. 2) Login as User: Method 2: Use a Modular Form. A: Increases the complexity of the app, and also the fact that the number of needed inputs in rather minimal.

small and data needed is rather minimal, it's wise to stick with traditional forms.

2) Create, Update, Edit and Delete Skills:

Method 1: Use Simple Forms and Buttons to perform CRUD.

A: Since the overall scale of the application is small and data needed is rather minimal, it's wise to stick with traditional forms.

Method 3: Use 3rd party services.

A: Valid method but due to the scale of the application, this method is seemingly overkill.

User Dashboard:

1) Create, Update, Edit and Delete Sessions:

Method 2: Have separate pages for each operation.

A: Since the user needs all information in one place and also that the number of operations they can do is limited, it is best to have a single page control centre type.

Method 3: Use Modals.

A: The modals pose a constant pop up like environment while performing CRUD operations.

2) Create, Update, Edit and Delete Skills:

Method 2: Have separate pages for each operation.

A: Since the user needs all information in one place and also that the number of operations they can do is limited, it is best to have a single page control centre type.

Method 3: Use Modals.

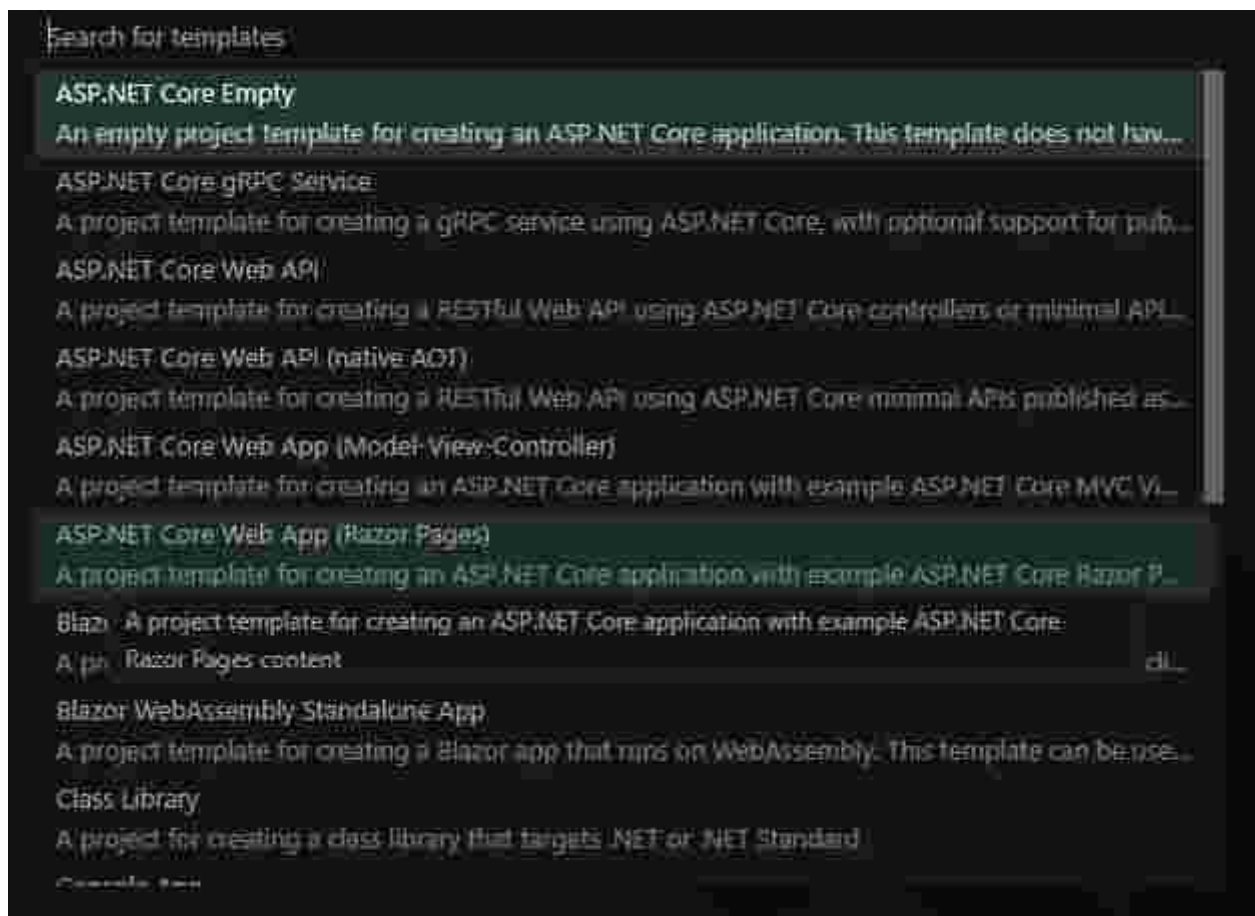
A: The modals pose a constant pop up like environment while performing CRUD operations.

Why?

Why Not?

Output Screenshots

Prepare the application...



Log into Azure Portal

Microsoft Azure



Pick an account

to continue to Microsoft Azure



KAVYA SRI VAIPAVA.S
21ucs099@kamarajengg-edu.in
Signed in

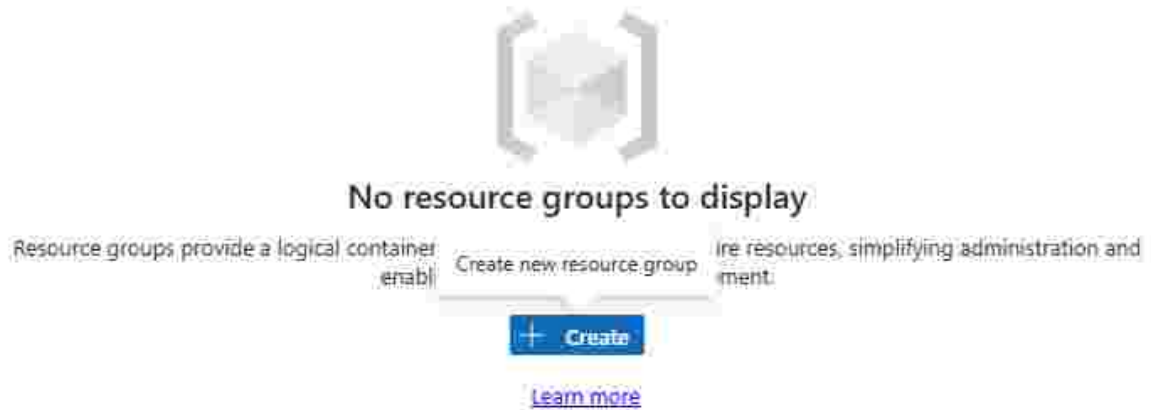


Nisanth Saravanan
nisanth.saravanan@relevantz.com
Connected to Windows



Use another account

Create a new resource group



Create a resource group ...

Basics Tags Review + create

Resource group - A container that holds related resources for an Azure solution. The resource group can include all the resources for the solution, or only those resources that you want to manage as a group. You decide how you want to allocate resources to resource groups based on what makes the most sense for your organization. [Learn more](#)

Subscription *	Azure for Students
Resource group name *	RixAssessment
Region *	(Asia Pacific) Central India

Resource groups

Kamaraj College of Engineering and Technology (kamarajengg.edu.in)



What are the different statuses available for resource groups?

Filter resources

[+ Create](#) [Manage view](#) [Refresh](#) [Export to CSV](#) [Open query](#) [Assign tags](#)

[You are viewing a new version of Browse experience. Click here to access the old experience.](#)

[Filter for any field...](#) [Subscription equals all](#) [Location equals all](#) [+ Add filter](#)



Na RixAssessment



[RixAssessment](#)



RixAssessment

Resource group



What are the best practices for managing this resource group?

Are there any alerts fired for this resource group?

How do I monitor this resource group?

[Search](#)

[+ Create](#) [Manage view](#) [Delete resource group](#) [Refresh](#) [Export to CSV](#) [Open query](#) [Assign tags](#) [Move](#) [Delete](#)

Overview

- [Activity log](#)
- [Access control \(IAM\)](#)
- [Tags](#)
- [Resource visualizer](#)
- [Events](#)
- [Settings](#)
- [Cost Management](#)
- [Monitoring](#)
- [Automation](#)
- [Help](#)

Essentials

Subscription [\(move\)](#) : [Azure for Students](#)

Deployments : [No deployments](#)

Subscription ID : c4a32f24-1087-4852-a36d-f7b99274390d

Location : Central India

Tags [\(edit\)](#) : [Add tags](#)

Resources

Recommendations

[Filter for any field...](#) [Type equals all](#) [Location equals all](#) [+ Add filter](#)

Showing 0 to 0 of 0 records. ☐ Show hidden types

No grouping

Name ↑

Type ↑↓

Location ↑↓



No resources match your filters

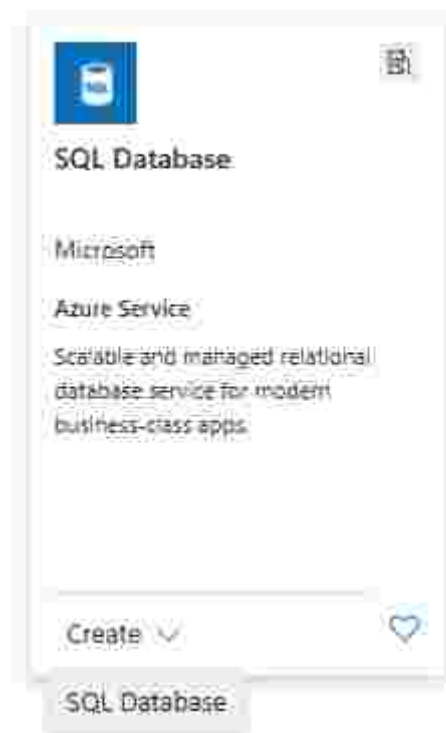
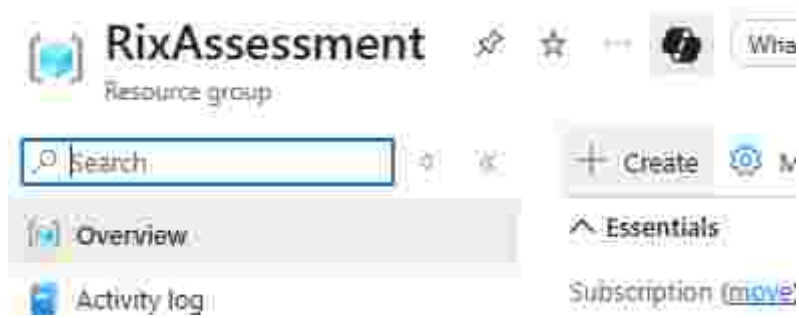
Try changing or clearing your filters.

[+ Create resources](#)

[Clear filters](#)

[Learn more](#)

Create a SQL server and db in the resource group



Create a SQL Server

Create SQL Database Server

Microsoft

Server details

Enter required settings for this server, including providing a name and location. This server will be created in the same subscription and resource group as your database.

Server name *	rixassessmentserver	✓
		.database.windows.net
Location *	(Asia Pacific) Central India	✓

Authentication

 Azure Active Directory (Azure AD) is now Microsoft Entra ID. [Learn more](#) 

Select your preferred authentication methods for accessing this server. Create a server admin login and password to access your server with SQL authentication; select only Microsoft Entra authentication [Learn more](#)  using an existing Microsoft Entra user, group, or application as Microsoft Entra admin [Learn more](#) , or select both SQL and Microsoft Entra authentication.

Authentication method	☐ Use Microsoft Entra-only authentication	
	☐ Use both SQL and Microsoft Entra authentication	
	☒ Use SQL authentication	
Server admin login *	nisanthrz	✓
Password *		✓
Confirm password *		✓

Create a SQL Database

Create SQL Database

Microsoft

Database details

Enter required settings for this database, including picking a logical server and configuring the compute and storage resources.

Database name *

Server *

[Create new](#)

Want to use SQL elastic pool? ☐ Yes ☒ No

Workload environment ☒ Development ☐ Production

i Default settings provided for Development workloads. Configurations can be modified as needed.

Compute + storage *

General Purpose - Serverless

Standard-series (Gen5), 1 vCore, 32 GB storage, zone redundant disabled

[Configure database](#)

Backup storage redundancy

Choose how your PITR and LTR backups are replicated. Geo restore or ability to recover from regional outage is only available when geo-redundant storage is selected.

Backup storage redundancy ☐ Locally-redundant backup storage ☐ Zone-redundant backup storage ☒ Geo-redundant backup storage ☐ Geo-Zone-redundant backup storage

Configuring the SQL Database

Create SQL Database

Microsoft

Basics **Networking** Security Additional settings Tags Review + create

Configure network access and connectivity for your server. The configuration selected below will apply to the selected server 'rxassessmentserver' and all databases it manages. [Learn more](#)

Network connectivity

Choose an option for configuring connectivity to your server via public endpoint or private endpoint. Choosing no access creates with defaults and you can configure connection method after server creation. [Learn more](#)

Connectivity method *

- ☐ No access
- ☒ Public endpoint
- ☐ Private endpoint

Firewall rules

Setting 'Allow Azure services and resources to access this server' to Yes allows communications from all resources inside the Azure boundary, that may or may not be part of your subscription. [Learn more](#)

Setting 'Add current client IP address' to Yes will add an entry for your client IP address to the server firewall.

Allow Azure services and resources to access this server *

No Yes

Add current client IP address *

No Yes

Review and Create

Create SQL Database

Microsoft

Basics Networking Security Additional settings Tags **Review + create**

Product details

SQL database
by Microsoft
[Terms of use](#) | [Privacy policy](#)

Estimated cost

Storage cost 5.45 USD / month + Compute cost 0.000159 USD / vCore second

Terms

By clicking "Create", I (a) agree to the legal terms and privacy statement(s) associated with the Marketplace offering(s) listed above; (b) authorize Microsoft to bill my current payment method for the fees associated with the offering(s), with the same billing frequency as my Azure subscription; and (c) agree that Microsoft may share my contact, usage and transactional information with the provider(s) of the offering(s) for support, billing and other transactional activities. Microsoft does not provide rights for third-party offerings. For additional details see [Azure Marketplace Terms](#).

Basics

Subscription	Azure for Students
Resource group	RixAssessment
Region	Central India
Database name	rixassessmentdb
Server	(new) rixassessmentserver
Authentication method	SQL authentication
Server admin login	nisanthr
Compute + storage	General Purpose - Serverless: Standard-series (Gen5), 1 vCore, 32 GB storage, zone redundant disabled
Backup storage redundancy	Geo-redundant backup storage



Cost summary

General Purpose (GP_S_Gen5_1)	
Cost per GB (in USD)	0.13
Max storage selected (in GB)	x 41.6
ESTIMATED STORAGE COST / MONTH	5.45 USD
COMPUTE COST / VCORE SECOND	0.000159 USD

NOTES

1 Serverless databases are billed in vCore seconds based on a combination of CPU and memory utilization. [Learn more about serverless billing](#)

Deployment is in progress

Deployment name : Microsoft.SQLDatabase.newDatabaseNewServer_b1c7afacf1174985b7192
Subscription : [Azure for Students](#)
Resource group : [RixAssessment](#)

Start time : 01/09/2025, 14:36:11
Correlation ID : a03443ca-7b47-4c48-b83c-79cdb797a5a7

Deployment details

Resource	Type	Status
rixassessmentserver	Microsoft.Sql/servers	Accepted
Resource	Type	Status
rixassessmentserver/rixassessmentdb	Microsoft.Sql/servers/databases	Created
rixassessmentserver/Default	Microsoft.Sql/servers/connectionpools	OK
rixassessmentserver/AllowAllWindowsAzureIps	Microsoft.Sql/servers/firewallrules	OK
rixassessmentserver/ClientIp-2025-9-1_14-36-5	Microsoft.Sql/servers/firewallrules	OK
rixassessmentserver	Microsoft.Sql/servers	Created

Go to the resource

rixassessmentdb (rixassessmentserver/rixassessmentdb)

SQL database

Search

Copy Restore Export Set server firewall Delete Connect with... Feedback

Overview

Activity log

Tags

Diagnose and solve problems

Query editor (preview)

Mirror database in Fabric (preview)

Resource visualizer

Settings

Compute + storage

Connection strings

Properties

Locks

Data management

Integrations

Power Platform

Security

Auditing

Ledger

Data Discovery & Classification

Dynamic Data Masking

Microsoft Defender for Cloud

Summarize the properties for this SQL database. Diagnose login failure errors List the access control settings for this SQL database.

Mirror databases in Microsoft Fabric. Easily replicate your existing databases in Fabric, and help your team achieve streamlined ETL and operational analytics goals. [Learn more](#)

Essentials

Resource group (mcp)

Status

Location

Subscription (mcp)

Subscription ID

Tags (add)

Server name

Connection strings

Pricing tier

Auto-pause delay

Earliest restore point

JSON View

Getting started

Monitoring

Properties

Features

Notifications (0)

Integrations

Tutorials

Start working with your database

Connect to your database and start working with data with a few simple steps. [Learn more](#)

Configure access

Configure network access to your SQL server. [Learn more](#)

Configure

Connect to application

Use connection strings to connect to your SQL database from your applications and favorite tools.

See connection strings

Start developing

Work in your database by using tools to add, modify and query data. [Compare tools](#)

Open Azure Data Studio

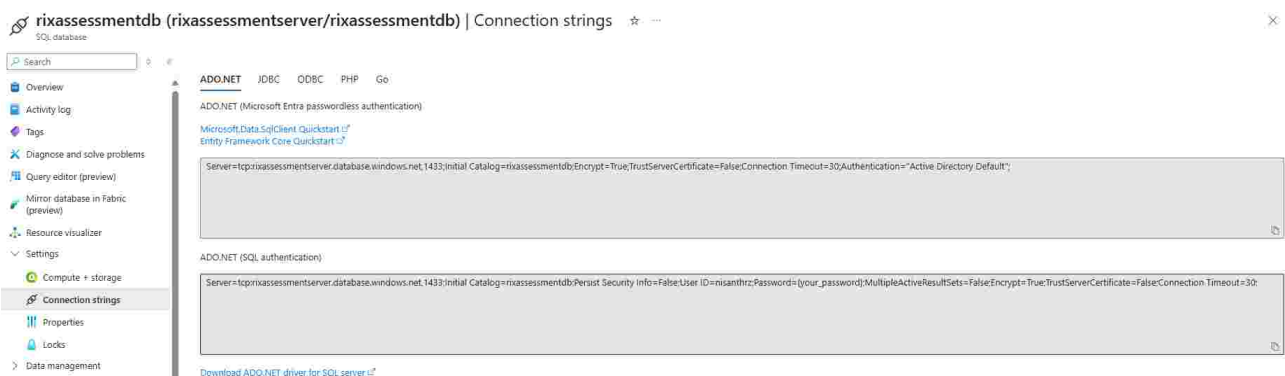
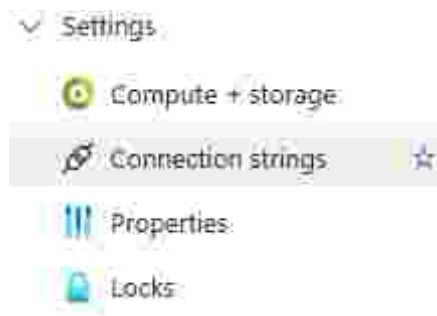
Open in Visual Studio

Open in Visual Studio Code

Mirror database in Fabric

Replicate existing databases in Fabric, and help your team achieve streamlined ETL and operational analytics goals. [Learn more](#)

Obtain Connection Strings



Copy the 2nd link and utilize it in the project for migrations...

```
1  [ ]
2  "logging": {
3    "LogLevel": {
4      "Default": "Information",
5      "Microsoft.AspNetCore": "Warning"
6    }
7  },
8  "ConnectionStrings": {
9    "DefaultConnection": "Server=tcp:rixassessmentserver.database.windows.net,1433;Initial Catalog=rixassessmentdb;Persist Security Info=False;User ID=rixanthrz;Password={your_password};MultipleActiveResultSets=False;Encrypt=True;TrustServerCertificate=False;Connection Timeout=30;"
10  },
11  "AllowedHosts": "*"
12  [ ]
```

Migrate and update Database

Migrations Successful

```
PS D:\Repos\RelevantzRIX2024\Intern Files\Intern Assessment\L1 Assessment\Azure\SkillTransferApp> dotnet ef migrations add Init
Build started...
Build succeeded.
Done. To undo this action, use 'ef migrations remove'
```

Database Update Failed

```
Error Number:40615,State:1,Class:14
Cannot open server 'rixassessmentserver' requested by the login. Client with IP address '47.247.170.166' is not allowed to access the server. To enable
e on the master database to create a firewall rule for this IP address or address range. It may take up to five minutes for this change to take effect.
```

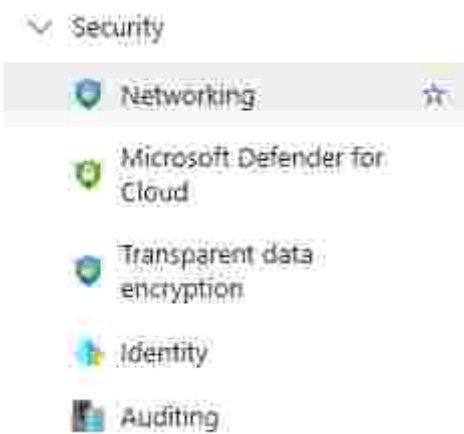
To resolve this error, Go to the resource on Azure Portal

The screenshot displays the Azure Portal interface for the 'rixassessmentdb' SQL database. The left-hand navigation pane includes sections for Overview, Activity log, Tags, Diagnose and solve problems, Query editor (preview), Mirror database in Fabric (preview), Resource visualizer, Settings, Compute + storage, Connection strings, Properties, Locks, Data management, Integrations, Power Platform, and Security. The main content area features a 'Getting started' section with a 'Start working with your database' heading. Below this, four cards provide guidance: 'Configure access' (with a 'Configure' button), 'Connect to application' (with a 'See connection strings' button), 'Start developing' (with buttons for 'Open Azure Data Studio', 'Open in Visual Studio', and 'Open in Visual Studio Code'), and 'Mirror database in Fabric'. A table of 'Essentials' is also visible, listing details such as Resource group (rixassessment), Status (Online), Location (Central India), Subscription (Azure for Students), and Subscription ID (c4a32f24-1087-4852-a36d-f7b99274390d).

Navigate to the SQL Server

The screenshot shows the Azure Portal interface for the 'rixassessmentserver' SQL server. The left-hand navigation pane includes sections for Overview, Activity log, Access control (IAM), Tags, Quick start, Diagnose and solve problems, Resource visualizer, Settings, Microsoft Entra ID, SQL databases, SQL elastic pools, Properties, Locks, Data management, Security, Intelligent performance, Monitoring, Automation, and Help. The main content area features a 'Getting started' section with a 'Start working with your database' heading. Below this, four cards provide guidance: 'Configure access' (with a 'Configure' button), 'Connect to application' (with a 'See connection strings' button), 'Start developing' (with buttons for 'Open Azure Data Studio', 'Open in Visual Studio', and 'Open in Visual Studio Code'), and 'Mirror database in Fabric'. A table of 'Essentials' is also visible, listing details such as Resource group (rixassessment), Status (Available), Location (Central India), Subscription (Azure for Students), and Subscription ID (c4a32f24-1087-4852-a36d-f7b99274390d).

Move to Networking Settings user Security



rixassessmentsserver | Networking

SQL server

Search

Feedback

Overview

Activity log

Access control (IAM)

Tags

Quick start

Diagnose and solve problems

Resource visualizer

Settings

Microsoft Entra ID

SQL databases

SQL elastic pools

Properties

Locks

Data management

Security

Networking

Microsoft Defender for Cloud

Transparent data encryption

Identity

Auditing

Public access

Private access

Connectivity

Public network access

Public Endpoints allow access to this resource through the internet using a public IP address. An application or resource that is granted access with the following network rules still requires proper authorization to access this resource. [Learn more](#)

Public network access

Disable

Selected networks

Connections from the IP addresses configured in the Firewall rules section below will have access to this database. By default, no public IP addresses are allowed. [Learn more](#)

Virtual networks

Allow virtual networks to connect to your resource using service endpoints. [Learn more](#)

+ Add a virtual network rule:

Rule	Virtual network	Subnet	Address range	Endpoint status	Resource group	Subscription	State
------	-----------------	--------	---------------	-----------------	----------------	--------------	-------

Firewall rules

Allow certain public internet IP addresses to access your resource. [Learn more](#)

+ Add your client IPv4 address (163.116.195.114) + Add a firewall rule.

Rule name	Start IPv4 address	End IPv4 address
Clientip-2025-9-1_14-36-5	163.116.195.114	163.116.195.114

Exceptions

Add a new firewall rule under Public Access

Firewall rules

Allow certain public internet IP addresses to access your resource. [Learn more](#)

+ Add your client IPv4 address (163.116.195.114) + Add a firewall rule

Provide IP range to cover all possible IPv4 Addresses, just this once

Add a firewall rule

Rule name	Start IP	End IP
CoverAll	0.0.0.0	255.255.255.255
		OK Cancel

Save

Firewall rules

Allow certain public internet IP addresses to access your resource. [Learn more](#)

+ Add your client IPv4 address (163.116.195.114) + Add a firewall rule

Rule name	Start IPv4 address	End IPv4 address	
CoverAll	0.0.0.0	255.255.255.255	
ClientIp-2025-9-1_14-36-5	163.116.195.114	163.116.195.114	

Exceptions

Retry DB Update

Success

```
PS D:\Repos\RelevantzRDX2024\Intern Files\Intern Assessment\L1 Assessment\Azure\SkillTransferApp> dotnet ef database update
Build started...
Build succeeded.
info: Microsoft.EntityFrameworkCore.Database.Command[20101]
      Executed DbCommand (44ms) [Parameters=[], CommandType='Text', CommandTimeout='30']
      SELECT 1
info: Microsoft.EntityFrameworkCore.Migrations[20411]
      Acquiring an exclusive lock for migration application. See https://aka.ms/efcore-docs-migrations-lock for more information if this takes too long.
Acquiring an exclusive lock for migration application. See https://aka.ms/efcore-docs-migrations-lock for more information if this takes too long.
info: Microsoft.EntityFrameworkCore.Database.Command[20101]
      Executed DbCommand (58ms) [Parameters=[], CommandType='Text', CommandTimeout='30']
      DECLARE @result int;
      EXEC @result = sp_getapplock @Resource = '__EFMigrationsLock', @LockOwner = 'Session', @LockMode = 'Exclusive';
      SELECT @result
info: Microsoft.EntityFrameworkCore.Database.Command[20101]
      Executed DbCommand (34ms) [Parameters=[], CommandType='Text', CommandTimeout='30']
      IF OBJECT_ID(N'__EFMigrationsHistory') IS NULL
      BEGIN
          CREATE TABLE [__EFMigrationsHistory] (
              [MigrationId] nvarchar(150) NOT NULL,
              [ProductVersion] nvarchar(32) NOT NULL,
              CONSTRAINT [PK__EFMigrationsHistory] PRIMARY KEY ([MigrationId])
          );
      END;
info: Microsoft.EntityFrameworkCore.Database.Command[20101]
      Executed DbCommand (33ms) [Parameters=[], CommandType='Text', CommandTimeout='30']
      SELECT 1
info: Microsoft.EntityFrameworkCore.Database.Command[20101]
      Executed DbCommand (33ms) [Parameters=[], CommandType='Text', CommandTimeout='30']
      SELECT OBJECT_ID(N'__EFMigrationsHistory');
info: Microsoft.EntityFrameworkCore.Database.Command[20101]
      Executed DbCommand (47ms) [Parameters=[], CommandType='Text', CommandTimeout='30']
      SELECT [MigrationId], [ProductVersion]
      FROM [__EFMigrationsHistory]
      ORDER BY [MigrationId];
info: Microsoft.EntityFrameworkCore.Migrations[20402]
      Applying migration '20250901091231_Init'.
Applying migration '20250901091231_Init'.
info: Microsoft.EntityFrameworkCore.Database.Command[20101]
      Executed DbCommand (36ms) [Parameters=[], CommandType='Text', CommandTimeout='30']
      CREATE TABLE [Sessions] (
          [Id] int NOT NULL IDENTITY,
          [SessionName] nvarchar(max) NULL,
          [SkillName] nvarchar(max) NULL,
          [DateTime] datetime2 NOT NULL,
          CONSTRAINT [PK_Sessions] PRIMARY KEY ([Id])
      );
info: Microsoft.EntityFrameworkCore.Database.Command[20101]
      Executed DbCommand (35ms) [Parameters=[], CommandType='Text', CommandTimeout='30']
      CREATE TABLE [Skills] (
          [Id] int NOT NULL IDENTITY,
          [StringName] nvarchar(max) NULL,
          [Proficiency] int NOT NULL,
          CONSTRAINT [PK_Skills] PRIMARY KEY ([Id])
      );
```

Creating a BLOB Storage

Navigate to the resource group

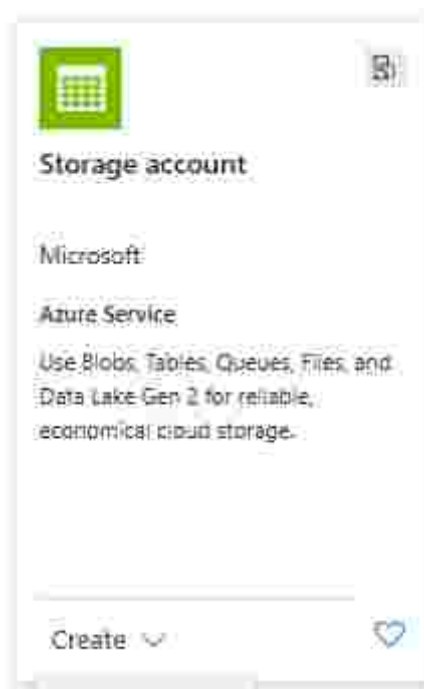
The screenshot shows the Azure portal interface for the 'RixAssessment' resource group. The left sidebar contains a navigation menu with options: Overview (selected), Activity log, Access control (IAM), Tags, Resource visualizer, Events, Settings, Cost Management, Monitoring, Automation, and Help. The main content area displays the 'Essentials' section with the following information:

- Subscription (move): [Azure for Students](#)
- Subscription ID: c4a32f24-1087-4652-a36d-f7b99274390d
- Tags (edit): [Add tags](#)

Below the Essentials section, there are tabs for 'Resources' and 'Recommendations'. The 'Resources' tab is active, showing a list of resources. At the top of the resources list, there is a filter bar with the text 'Filter for any field...' and two filter buttons: 'Type equals all' and 'Location equals a'. Below the filter bar, it says 'Showing 1 to 2 of 2 records' and a checkbox for 'Show hidden types'. The resources listed are:

- ☐ [Name ↑](#)
- ☐ [rixassessmentdb \(rixassessmentserver/rixassessmentdb\)](#)
- ☐ [rixassessmentserver](#)

Create a new storage account



Create a storage account

[Details](#) [Subscription](#) [Region](#) [Create a storage account](#) [Encryption](#) [Tags](#) [Review](#) [Create](#)

Azure Storage is a Microsoft-managed service providing cloud storage that is highly available, secure, durable, scalable, and redundant. Azure Storage includes Azure Blobs (objects), Azure Data Lake Storage Gen2, Azure Files, Azure Queues, and Azure Tables. The cost of your storage account depends on the usage and the options you choose below. [Learn more about Azure storage accounts](#)

Project details

Select the subscription in which to create the new storage account. Choose a new or existing resource group to organize and manage your storage account together with other resources.

Subscription *	Azure for Students
Resource group *	RixAssessment Create new

Instance details

Storage account name * ^①	rixassessmentsstorage
Region * ^②	(Asia Pacific) Central India Deploy to an Azure Extended Zone
Preferred storage type	Choose preferred storage type Azure Blob Storage or Azure Data Lake Storage Gen 2 Azure Files Other (tables and queues)
Performance * ^③	☐ Premium: Recommended for scenarios that require low latency.

Review and create

✔ Your deployment is complete

Deployment name: rixassessmentstorage_1756719423826 Start time: 01/09/2025, 15:07:39
Subscription: [Azure for Students](#) Correlation ID: 63ead2fa-84c0-4784-a913-ae78980db9d1
Resource group: RixAssessment

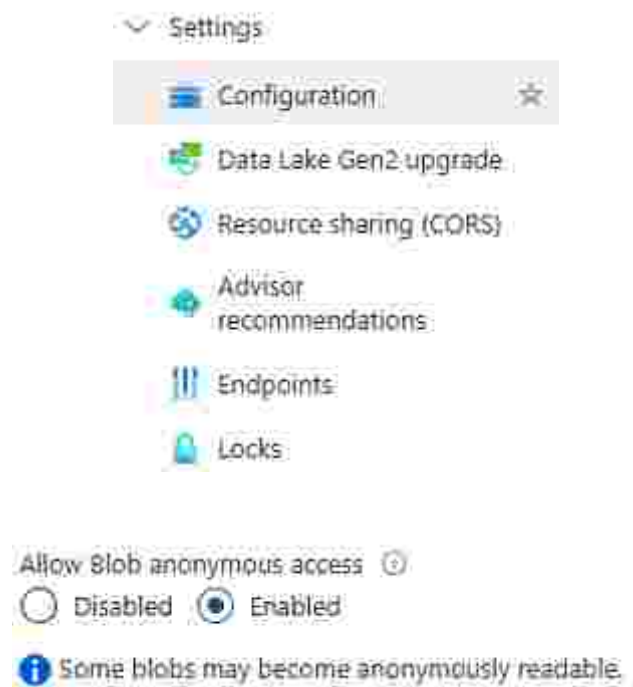
Deployment details

Resource	Type	Status
 rixassessmentstorage/default	Microsoft.Storage/storageAccounts/blobServices	OK
 rixassessmentstorage	Microsoft.Storage/storageAccounts	OK

Go to resource

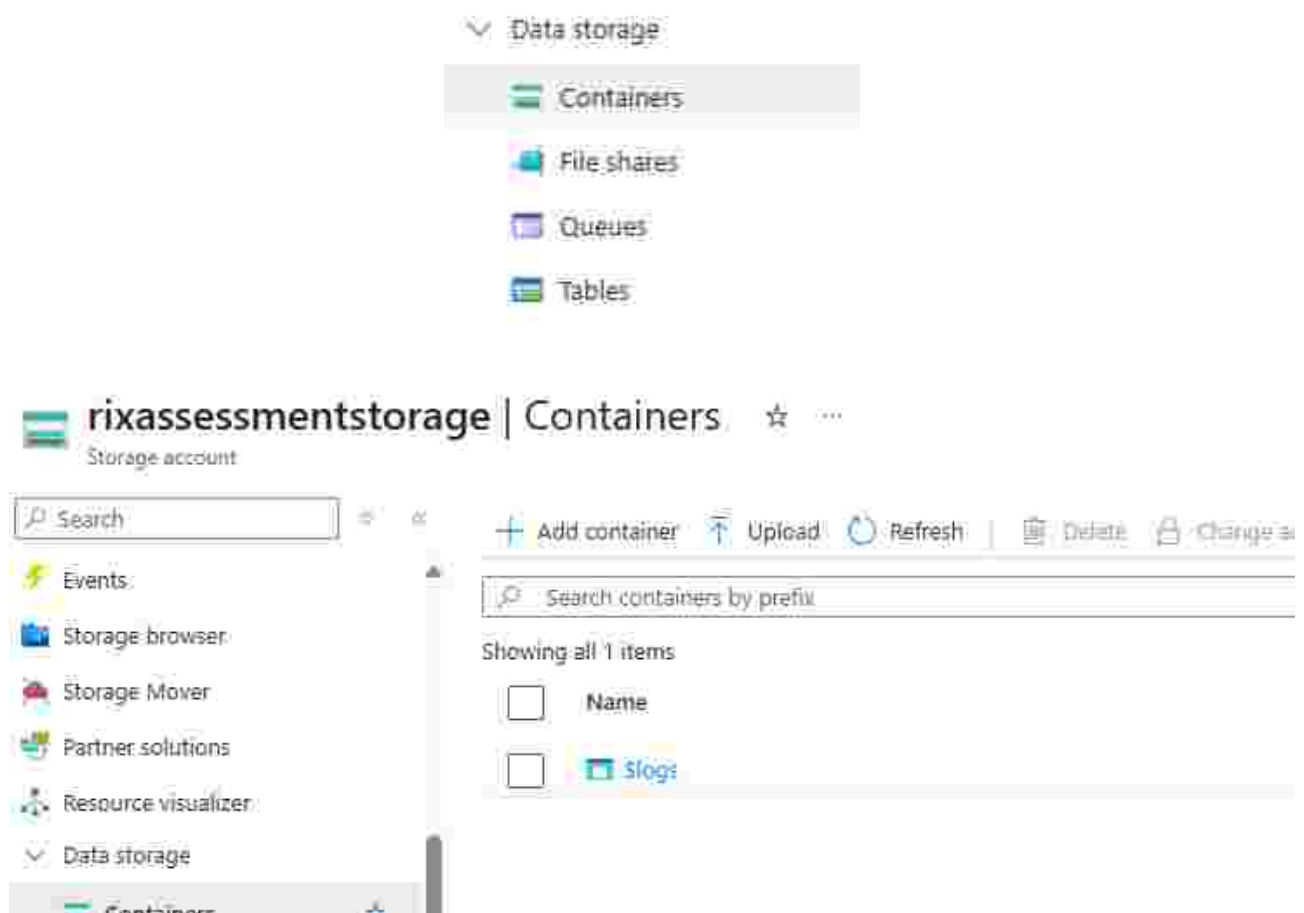
The screenshot displays the Azure portal interface for a storage account named 'RixAssessmentStorage'. The left-hand navigation pane is visible, with 'Data storage' selected. The main content area shows the 'Essentials' section, which provides a summary of the storage account's configuration. Key details include the resource group 'RixAssessment', the location 'centralindia', the subscription 'Azure for Students', and the disk state 'Primary: Available, Secondary: Available'. Below the essentials, the 'Properties' section is expanded, showing the 'Blob service' is currently 'Disabled'. The top of the page features a search bar and a toolbar with various actions like 'Upload', 'Open in Explorer', 'Delete', 'Move', 'Refresh', 'Open in mobile', 'CLI / PS', and 'Feedback'.

Enable Anonymous Access



Save Changes

Navigate to containers and create a new container



New container

Name *

assessment-container

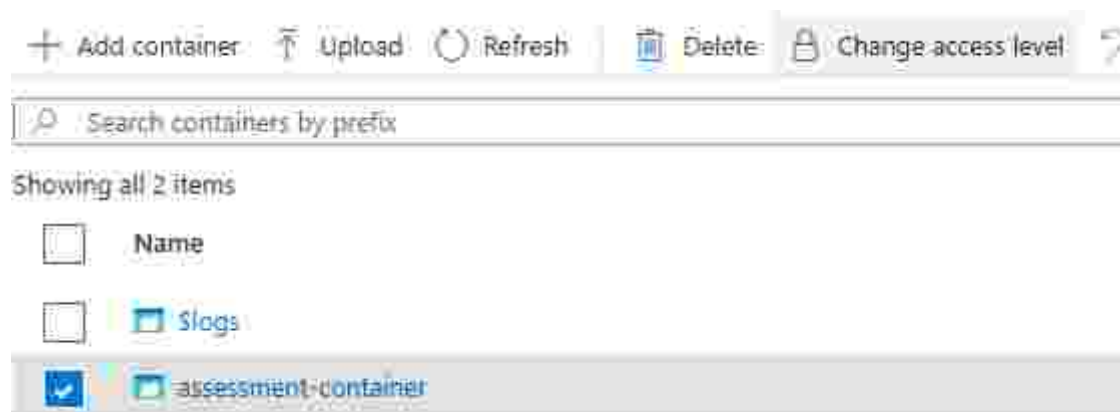
Anonymous access level ⓘ

Private (no anonymous access)

i The access level is set to private because anonymous access is disabled on this storage account.

Advanced

Change access level



Change access level

Change the access level of container "assessment-container".

Anonymous access level ⓘ

Blob (anonymous read access for blobs only)

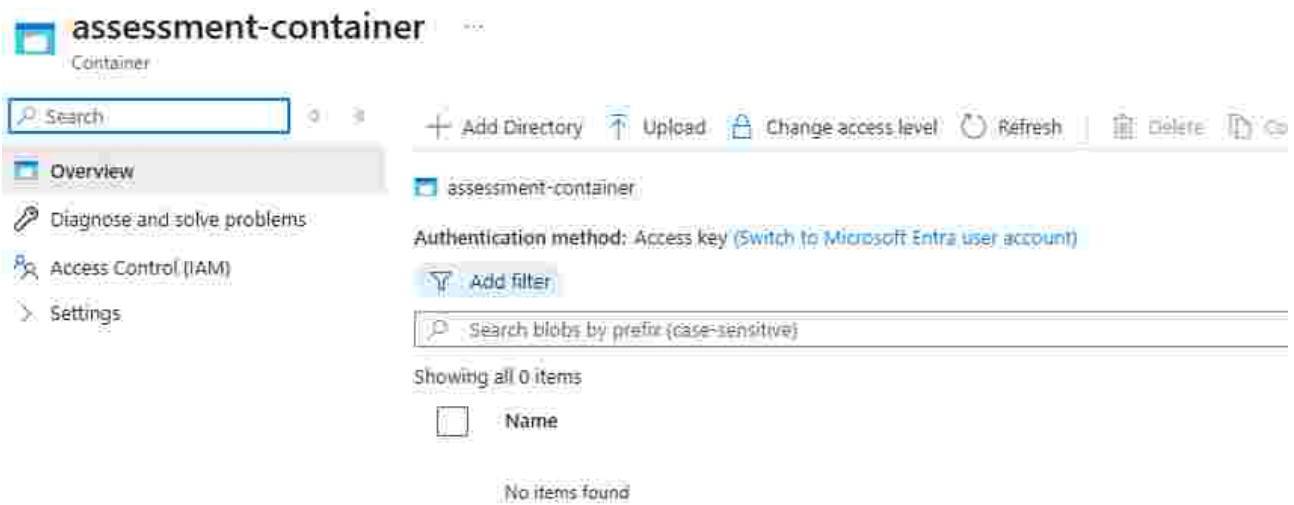
Private (no anonymous access)

Blob (anonymous read access for blobs only)

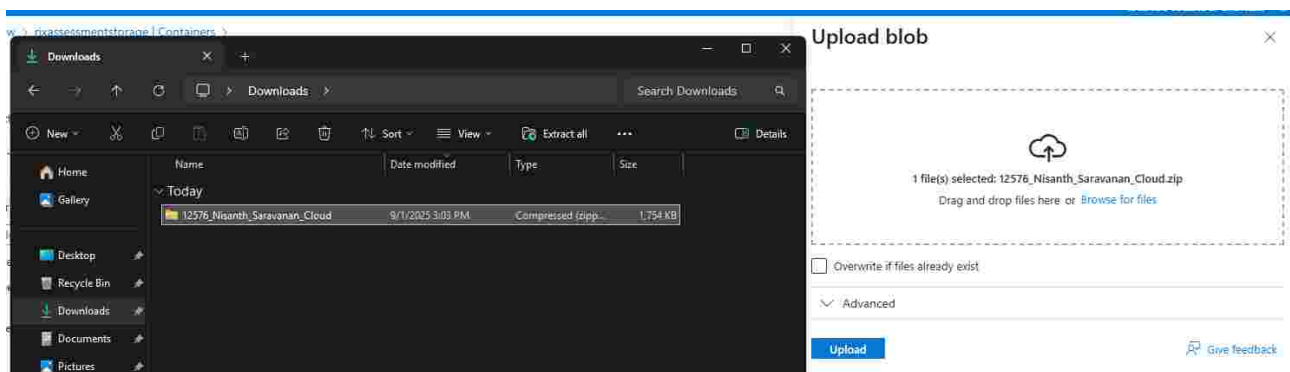
Container (anonymous read access for containers and blobs)

Save Changes

Open the Container



ZIP your project and upload it into the BLOG Storage container



 **assessment-container** Container

Search ⚙️ 🔍 + Add Directory ⬆️ Upload 🔒 Change access level 🔄 Refresh 🗑️

Overview

- 🔧 Diagnose and solve problems
- 👤 Access Control (IAM)
- > Settings

 **assessment-container**

Authentication method: Access key ([Switch to Microsoft Entra user account](#))

🔍 Add filter:

Search blobs by prefix (case-sensitive)

Showing all 1 items

☐	Name
☐	 12576_Nisanth_Saravanan_Cloud.zip

Open the uploaded file to get the download link for later...

12576_Nisanth_Saravanan_Cloud.zip

Blob

 Save  Discard  Download  Refresh  Delete  Change tier  Acquire lease  Break lease  Give feedback

Overview Versions Snapshots Edit Generate SAS

Properties

URL

<https://rxassessmentst...>

Copy to clipboard

LAST MODIFIED

01/09/2025, 3:14:06 pm

CREATION TIME

01/09/2025, 3:14:06 pm

VERSION ID

-

TYPE

Block blob

Create a VM

Navigate to resource group

RixAssessment
Resource group

Search

Overview

- Activity log
- Access control (IAM)
- Tags
- Resource visualizer
- Events
- Settings
- Cost Management
- Monitoring
- Automation
- Help

How do I monitor this resource group? What are the best practices for managing this resource group? How do I troubleshoot issues w

Create Manage view Delete resource group Refresh Export to CSV Open query Assign tags

Essentials

Subscription (move): [Azure for Students](#) Deployments: [2 Succeeded](#)

Subscription ID: c4a32f24-1087-4852-a36d-f7b99274390d Location: Central India

Tags (edit): [Add tags](#)

Resources Recommendations

Filter for any field... Type equals all Location equals all Add filter

Showing 1 to 3 of 3 records. ☐ Show hidden types

Name ↑	Type ↑
rixassessmentdb (rixassessmentserver/rixassessmentdb)	SQL database
rixassessmentserver	SQL server
rixassessmentstorage	Storage account

Create a new VM

Virtual machine

Microsoft

Azure Service

Azure Virtual Machines provide on-demand, high-scale, secure and virtualized infrastructure using either Linux or Windows operating systems.

Create

Configuring the creation

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription *	Azure for Students
Resource group *	RixAssessment Create new

Instance details

Virtual machine name *	RixAssessmentMachine
Region *	(Asia Pacific) Central India Deploy to an Azure Extended Zone
Availability options	Availability zone
Zone options	☒ Self-selected zone Choose up to 3 availability zones, one VM per zone ☐ Azure-selected zone (Preview) Let Azure assign the best zone for your needs Using an Azure-selected zone is not supported in region 'Central India'.
Availability zone *	Zone 2 You can now select multiple zones. Selecting multiple zones will create one VM per zone. Learn more

Security type	Trusted launch virtual machines Configure security features
Image *	Windows Server 2025 Datacenter: Azure Edition - x64 Gen2 See all images Configure VM generation
VM architecture	☐ Arm64 ☒ x64 Arm64 is not supported with the selected image.
Run with Azure Spot discount	☐
Size *	Standard_D2s_v3 - 2 vcpus, 8 GiB memory (US\$143.81/month) See all sizes

Administrator account

Username *	rixanthrz
Password *	*****
Confirm password *	*****

Basics Disks Networking Management Monitoring Advanced Tags Review + create

Azure VMs have one operating system disk and a temporary disk for short-term storage. You can attach additional data disks. The size of the VM determines the type of storage you can use and the number of data disks allowed. [Learn more](#)

VM disk encryption

Azure disk storage encryption automatically encrypts your data stored on Azure-managed disks (OS and data disks) at rest by default when persisting it to the cloud.

Encryption at host



Encryption at host is not registered for the selected subscription. [Learn more](#)

OS disk

OS disk size

Image default (127 GiB)

OS disk type *

Standard SSD (locally-redundant storage)

The selected VM size supports premium disks. We recommend Premium SSD for high IOPS workloads. Virtual machines with Premium SSD disks qualify for the 99.9% connectivity SLA.

Delete with VM



Review and create

Create a virtual machine



Help me choose the right VM size for my workload

Help me create a low-cost VM

Validation passed

Basics Disks Networking Management Monitoring Advanced Tags Review + create

Price

1 X Standard D2s v3

by Microsoft

[Terms of use](#) | [Privacy policy](#)

Subscription credits apply

0.1970 USD/hr

[Pricing for other VM sizes](#)

TERMS

By clicking "Create", I (a) agree to the legal terms and privacy statement(s) associated with the Marketplace offering(s) listed above; (b) authorize Microsoft to bill my current payment method for the fees associated with the offering(s), with the same billing frequency as my Azure subscription; and (c) agree that Microsoft may share my contact, usage and transactional information with the provider(s) of the offering(s) for support, billing and other transactional activities. Microsoft does not provide rights for third-party offerings. See the [Azure Marketplace Terms](#) for additional details.

✓ Your deployment is complete



Deployment name: CreateVm-MicrosoftWindowsServer.WindowsSe...
Subscription: [Azure for Students](#)
Resource group: [RixAssessment](#)

Start time: 01/09/2025, 15:19:24
Correlation ID: 14097365-533e-4eef-847b-fa3ba1

Deployment details

Resource	Type	Status
✓ RixAssessmentMachine	Microsoft.Compute/virtualMachines	OK
✓ rixassessmentmachine675_z2	Microsoft.Network/networkInterfaces	Created
✓ RixAssessmentMachine-vnet	Microsoft.Network/virtualNetworks	OK
✓ RixAssessmentMachine-nsg	Microsoft.Network/networkSecurityGroups	OK
✓ RixAssessmentMachine-ip	Microsoft.Network/publicIpAddresses	OK

Go to resource



RixAssessmentMachine

Virtual machine



Help me copy this VM in any region

Manage this VM with Azure CLI

Search

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Connect

Connect

Bastion

Windows Admin Center

Networking

Network settings

Load balancing

⚠ RixAssessmentMachine virtual machine agent status is not ready. Troubleshoot the issue →

Connect

Start

Restart

Stop

Hibernate

Capture

Delete

Refresh

Open in mobile

Essentials

Resource group [\(move\)](#) : [RixAssessment](#)

Status : Running

Location : Central India (Zone 2)

Subscription [\(move\)](#) : [Azure for Students](#)

Subscription ID : c4a32f24-1087-4852-a36d-f7b99274390d

Availability zone : 2

Tags [\(edit\)](#) : [Add tags](#)

Properties

Monitoring

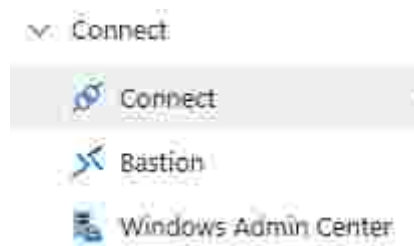
Capabilities (8)

Recommendations

Tutorials

Connection to our VM

Navigate to Connect



RixAssessmentMachine | Connect ☆ ...

Virtual machine

Search

Overview
Activity log
Access control (IAM)
Tags
Diagnose and solve problems
Resource visualizer
Connect
Connect
Bastion
Windows Admin Center
Networking
Network settings
Load balancing
Application security groups
Network manager
Settings
Availability + scale
Security

Now view, configure, and even save your connection settings—all in one place. Have comments or suggestions for our r

Refresh Reset password or keys Troubleshoot Feedback

Native RDP ⓘ MOST POPULAR LOCAL MACHINE

Source machine

Source machine OS ⓘ Windows

Source IP address ⓘ Local IP range | 163.116.0.0/16

Destination VM

VM IP address ⓘ Public IP | 20.40.41.207

VM port ⓘ 3389 ⓘ

Connection prerequisites

VM access ⓘ Check inbound NSG rules

Check access

Connect using RDP file

Download and open file to connect **Download RDP file**

Username nisanthrz ⓘ
Forgot password? [Reset password](#)

Edit settings ⓘ

More ways to connect (4)

Check access to native RDP port and download the RDP File

Connection prerequisites

VM access ⓘ

○ Checking inbound NSG rules...

Checking access...

i VM access check takes a minute. Continue on, but return here if you can't connect.

Connection prerequisites

VM access ⓘ

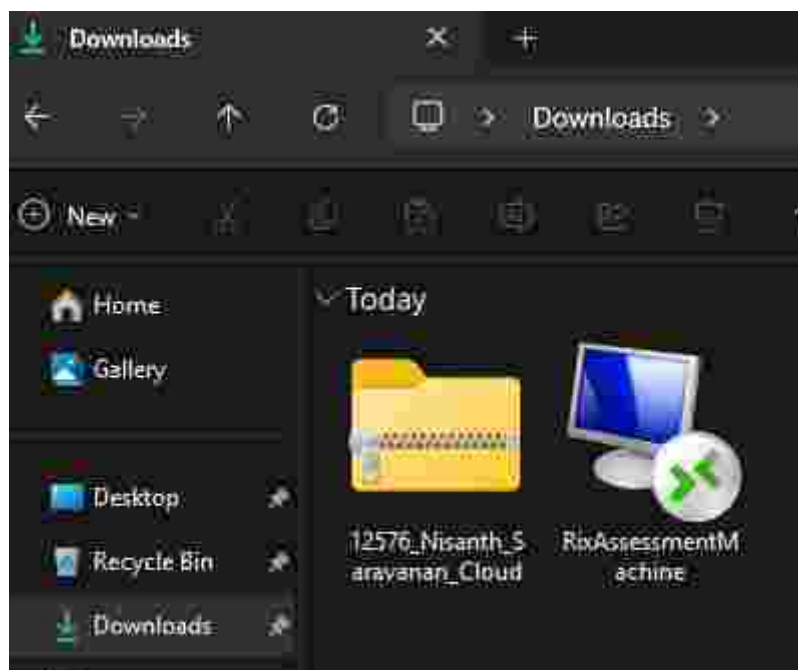
✔ Port 3389 is accessible from source IP(s) [View applied NSG rules](#)

Check access

Connect using RDP file

Download and open file to connect

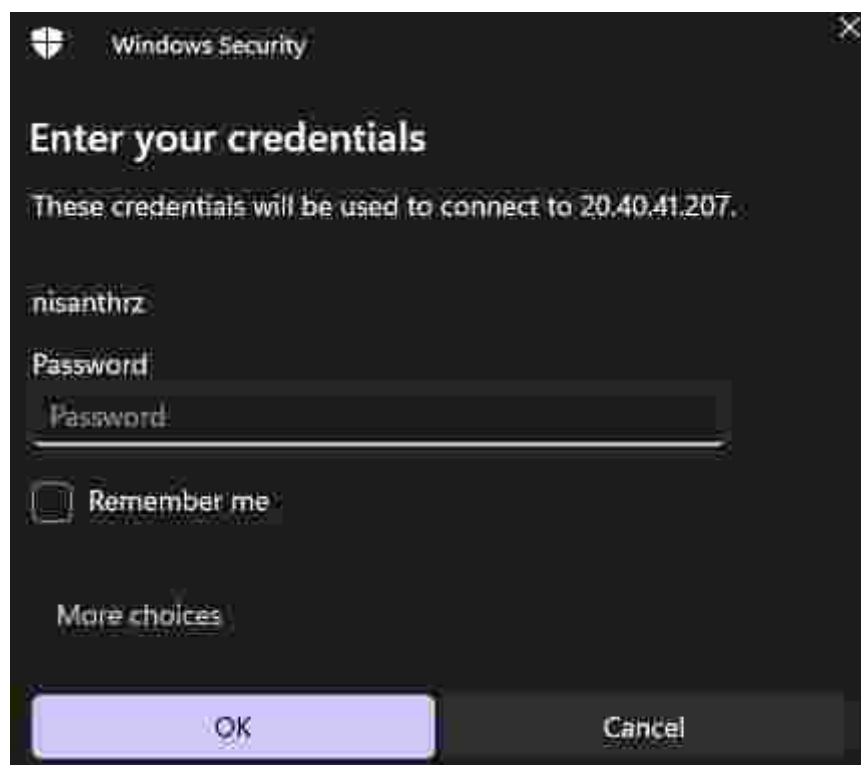
Download RDP file

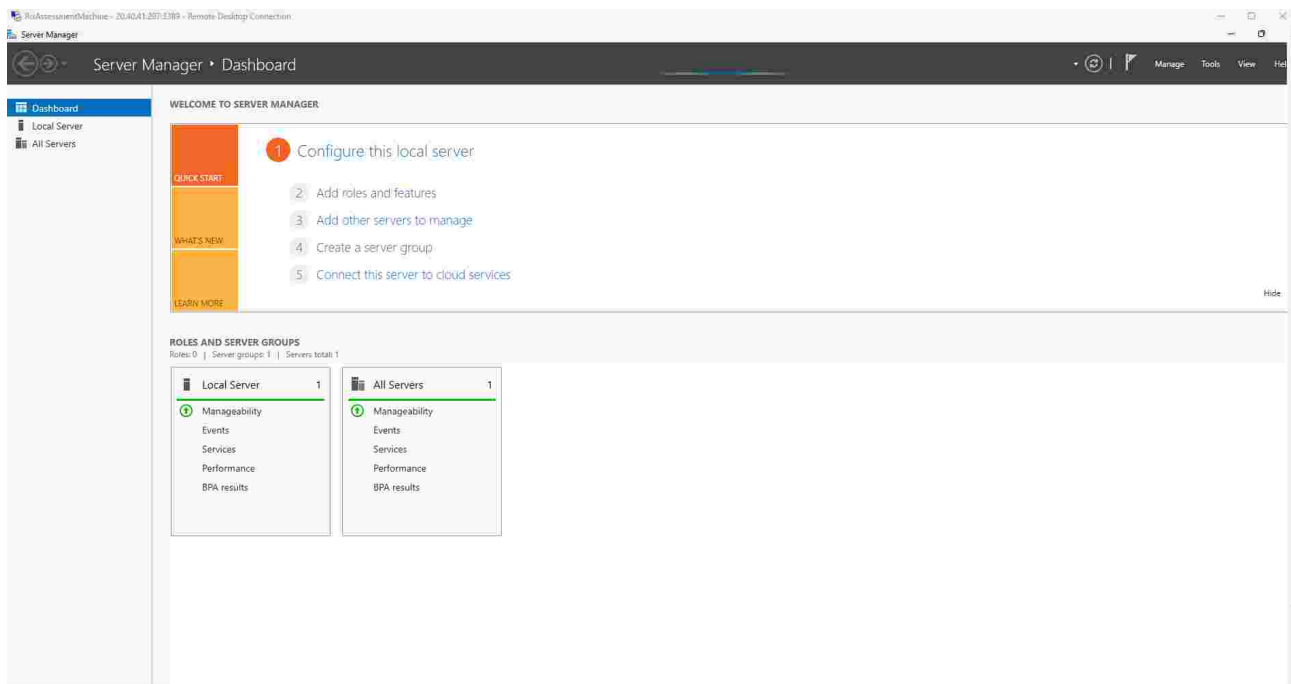


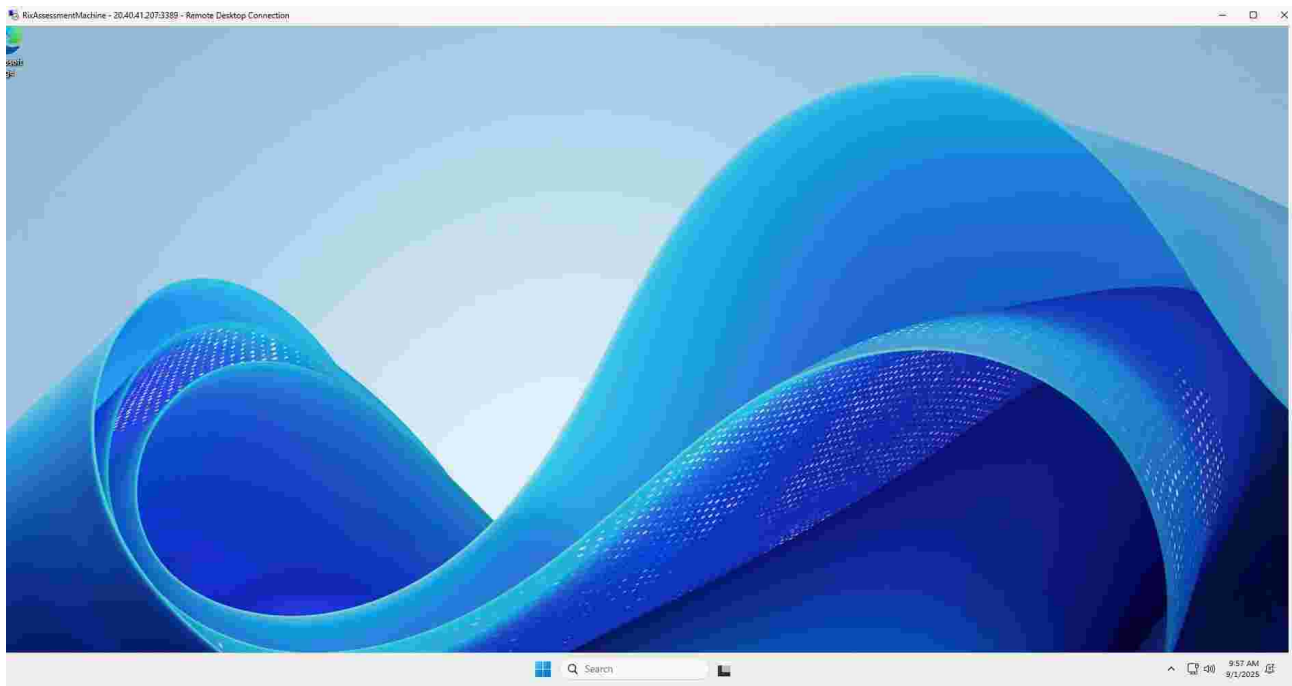
Open the RDP File



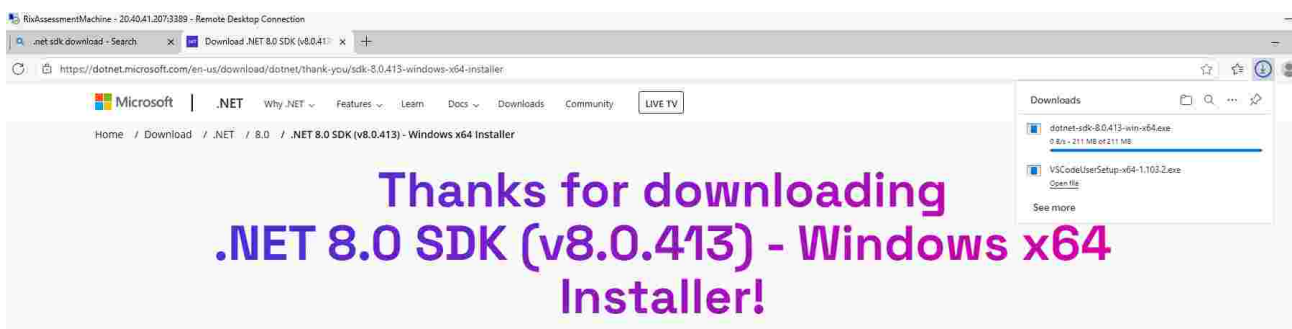
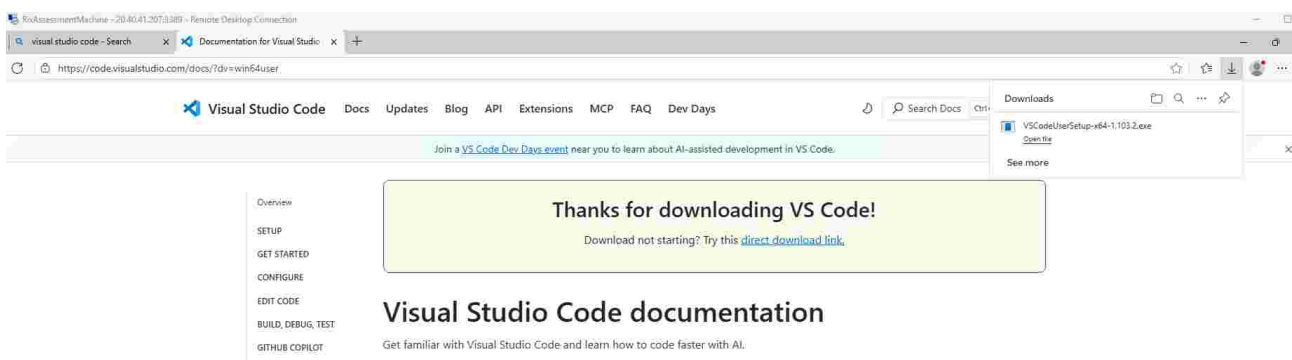
Enter the credentials



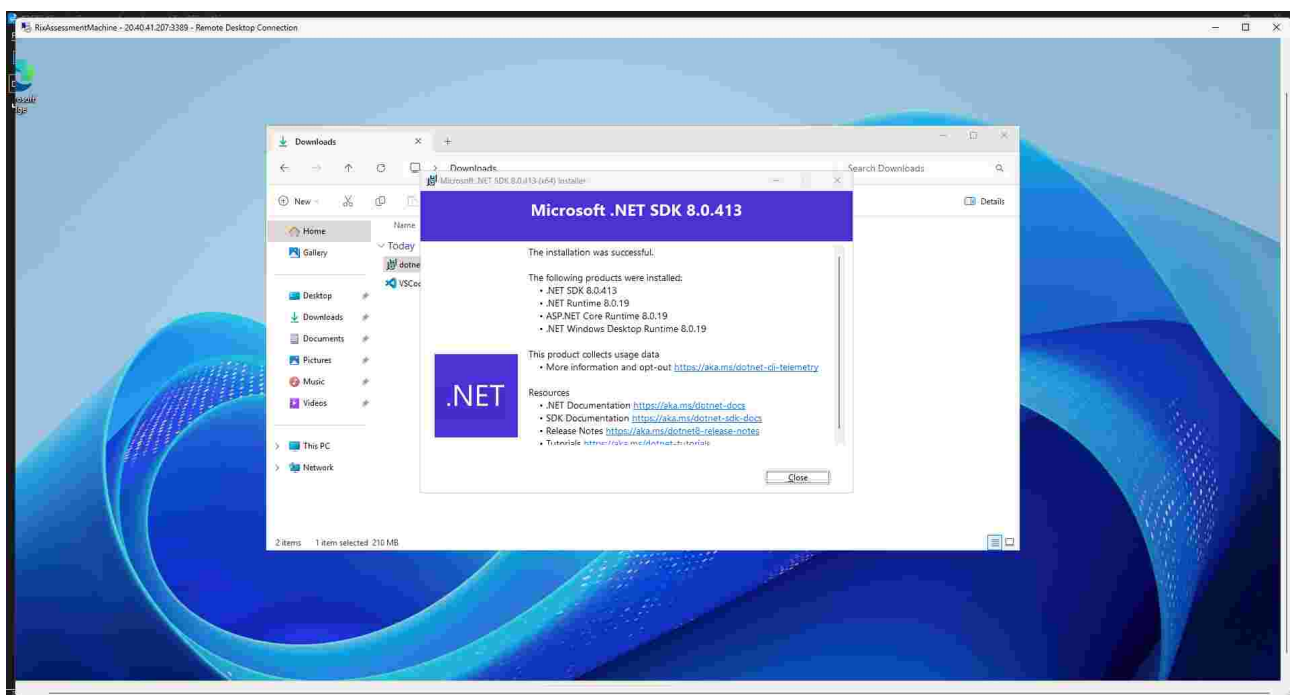
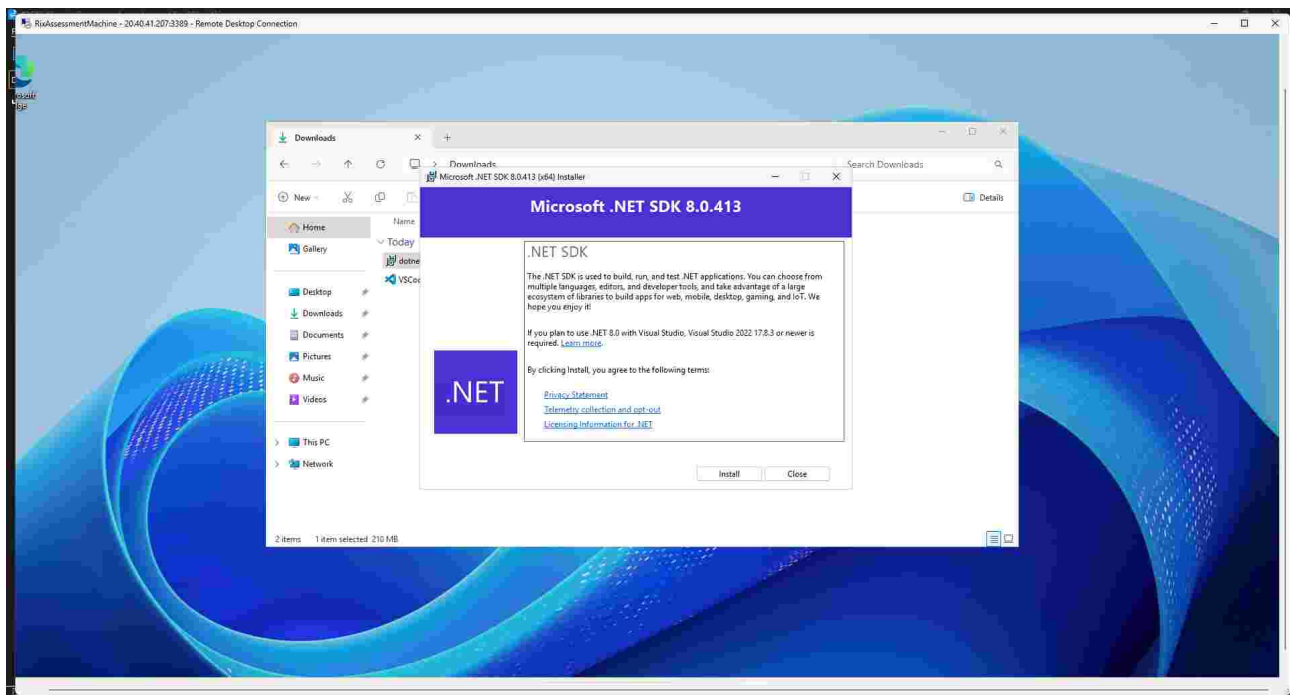


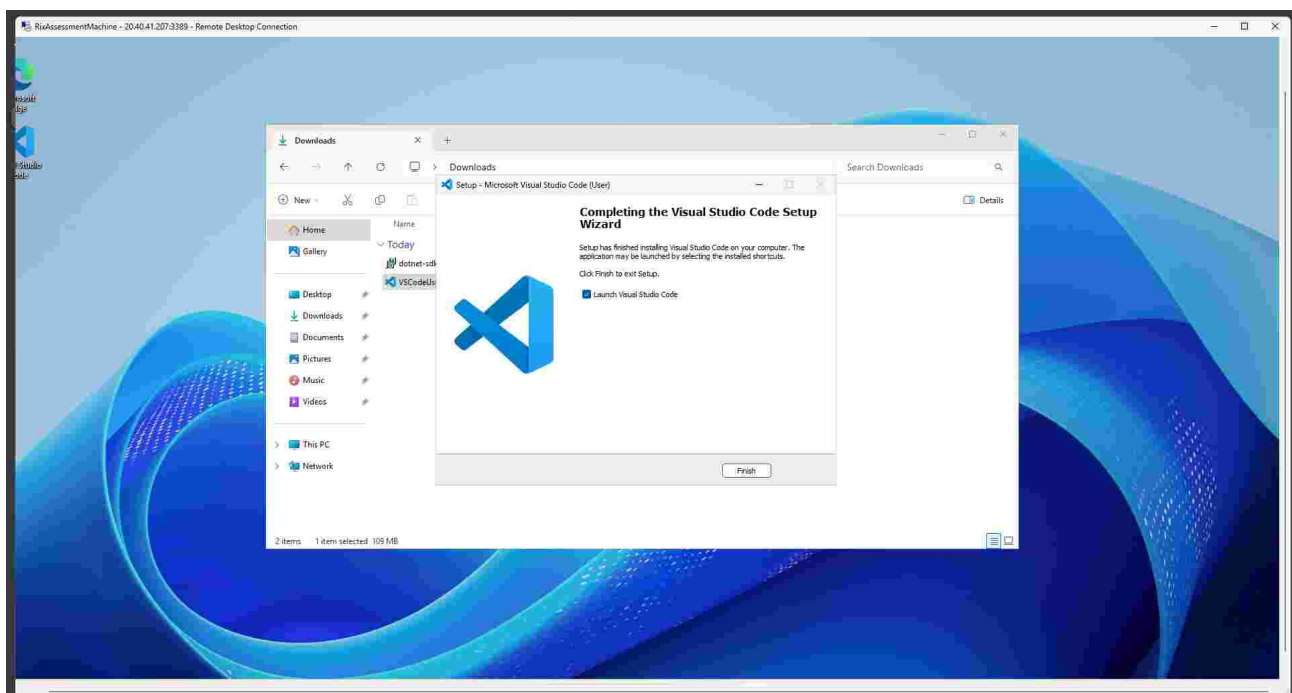
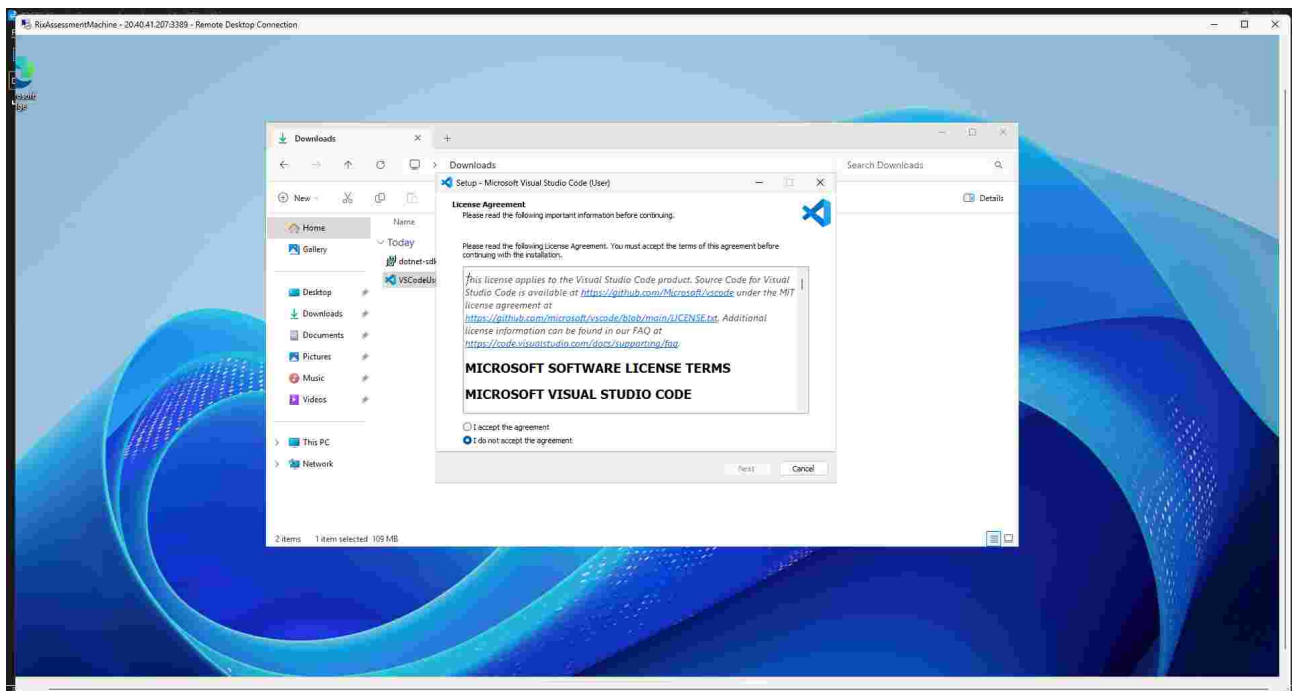


Download Visual Studio Code and .NET SDK



Install them both





Importing the project into the VM

Return to the BLOB Storage and navigate to out file that is already inside a container

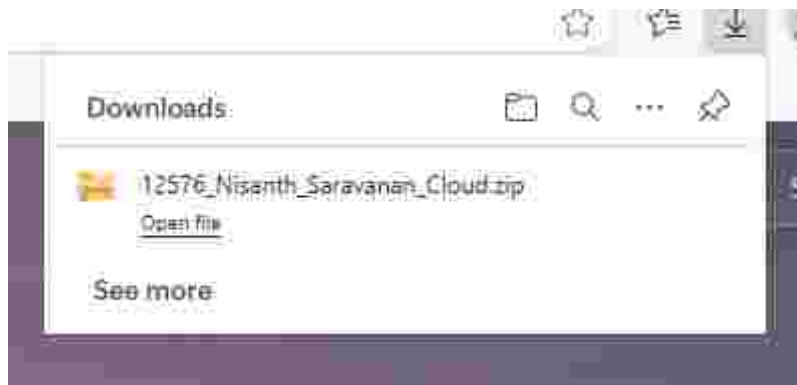
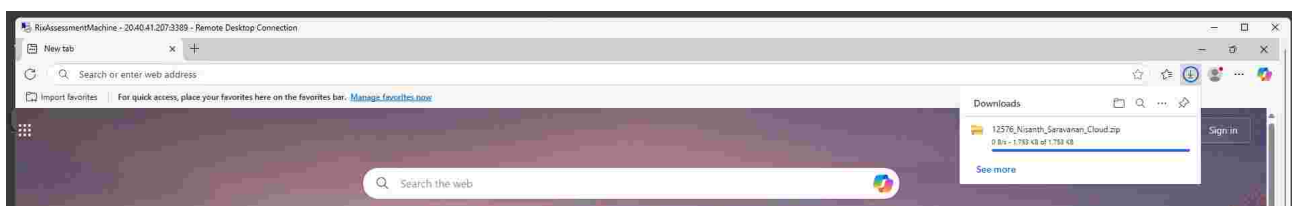
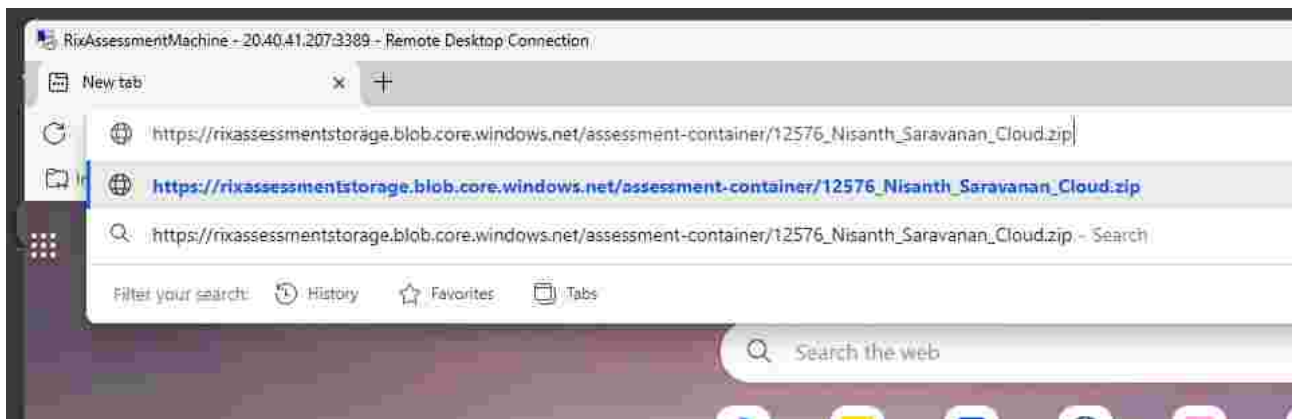
The screenshot shows the Azure portal interface. On the left, the 'assessment-container' is selected under 'Containers'. The main pane shows the '12576_Nisanth_Saravanan_Cloud.zip' blob. The 'Properties' tab is active, displaying the following details:

Property	Value
URL	https://rixassessmentstl...
LAST MODIFIED	01/09/2025, 3:14:06 pm
CREATION TIME	01/09/2025, 3:14:06 pm
VERSION ID	-
TYPE	Block blob
SIZE	1.71 MB
ACCESS TIER	Hot (inferred)
ACCESS TIER LAST MODIFIED	N/A
ARCHIVE STATUS	-
REHYDRATE PRIORITY	-
SERVER ENCRYPTED	true
ETAG	0x80DE93C17E072FC
VERSION-LEVEL IMMUTABILITY POLICY	Disabled
CACHE-CONTROL	
CONTENT-TYPE	application/x-zip-compressed
CONTENT-MD5	3aFVm8FszZWv3jopY8AFTw...
CONTENT-ENCODING	
CONTENT-LANGUAGE	
CONTENT-DISPOSITION	
LEASE STATUS	Unlocked
LEASE STATE	Available

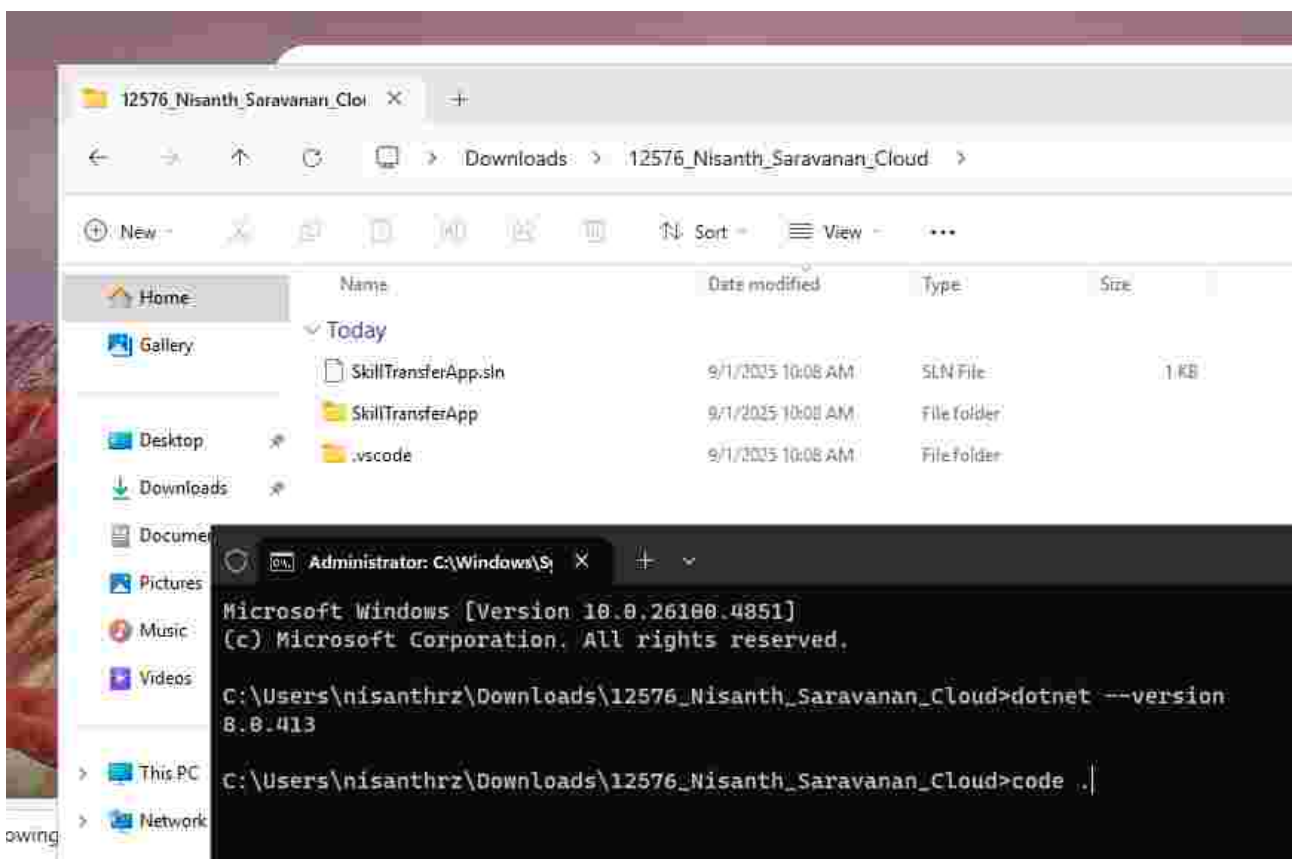
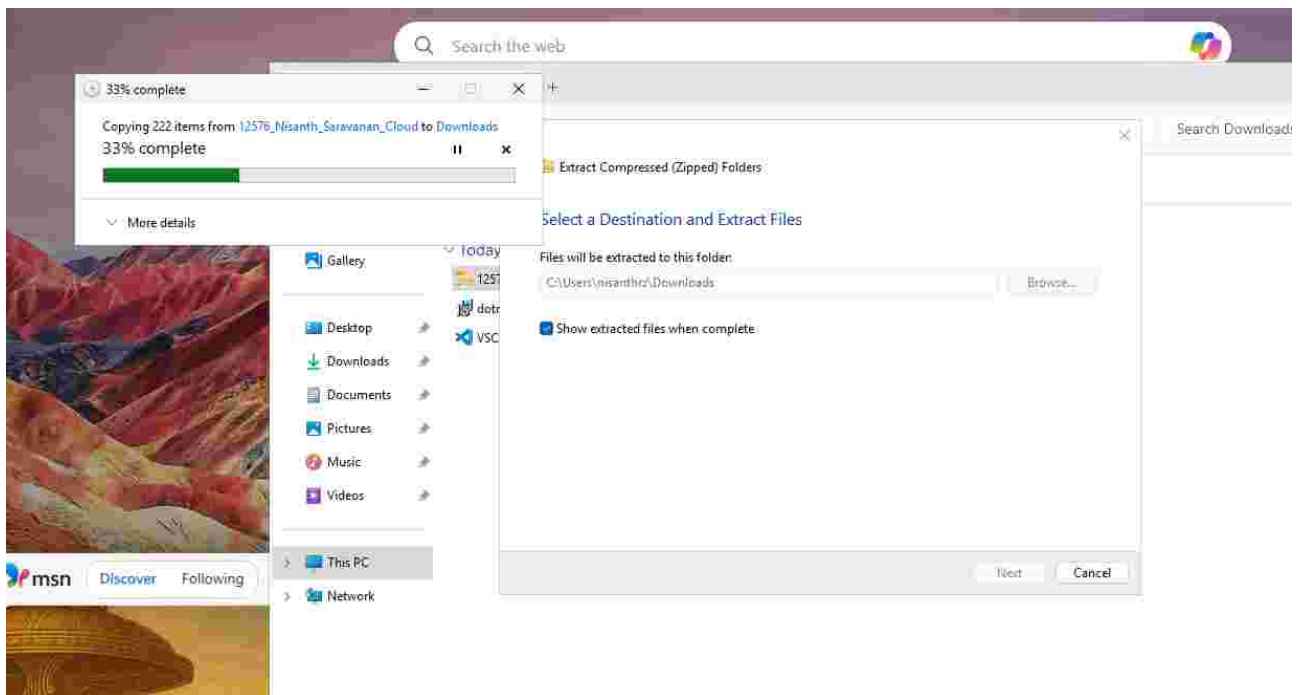
Copy the URL

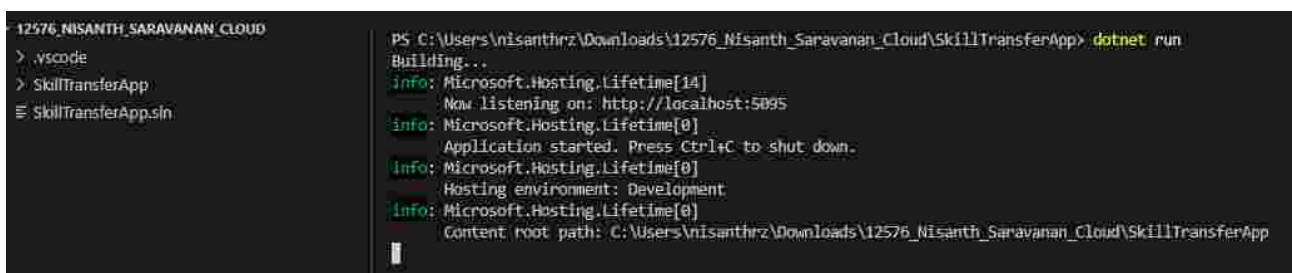
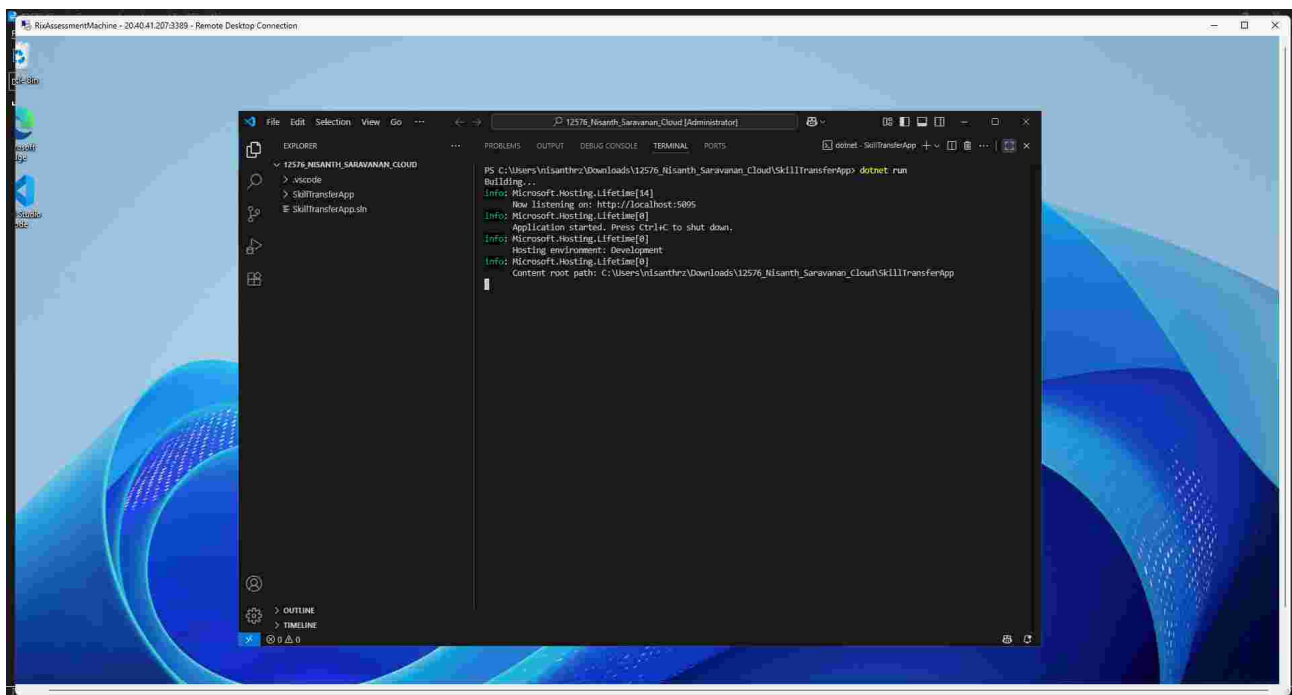
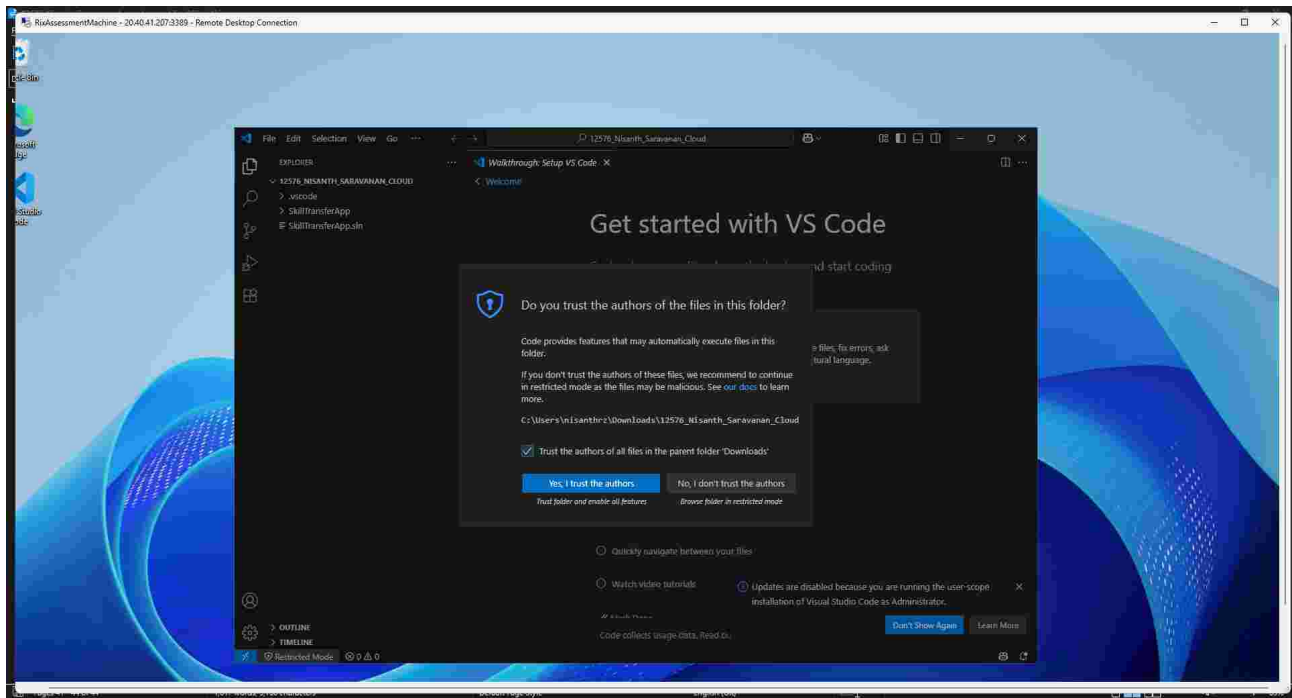
This screenshot is a zoomed-in view of the '12576_Nisanth_Saravanan_Cloud.zip' blob details. The 'Overview' tab is selected. The 'URL' property is highlighted, and a 'Copy to clipboard' tooltip is visible next to the URL text: <https://rixassessmentstl...>

Paste the URL in the address bar of the browser within our VM

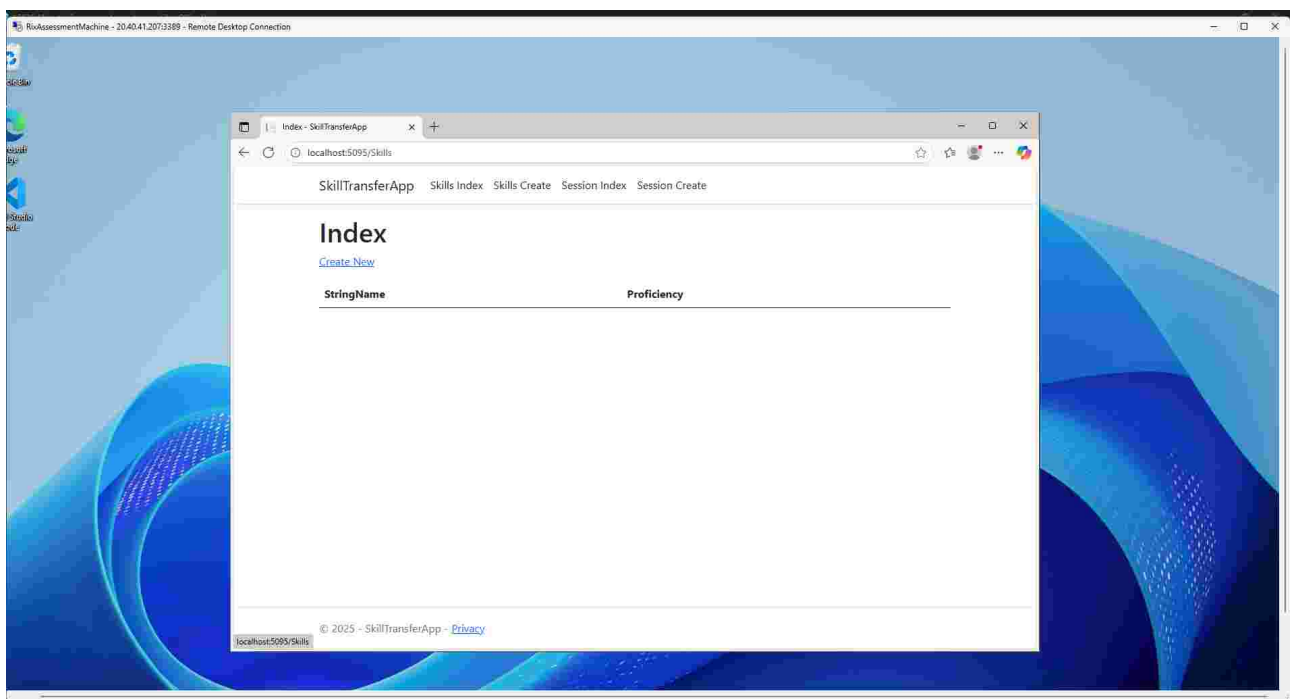
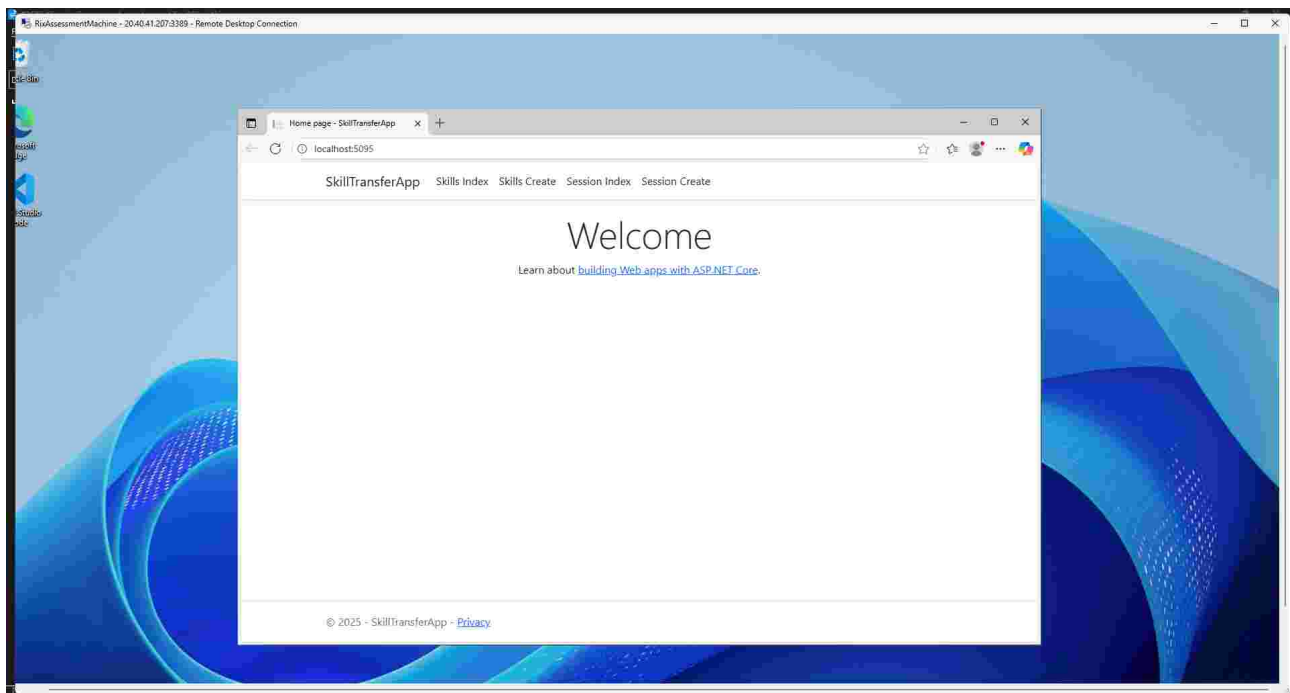


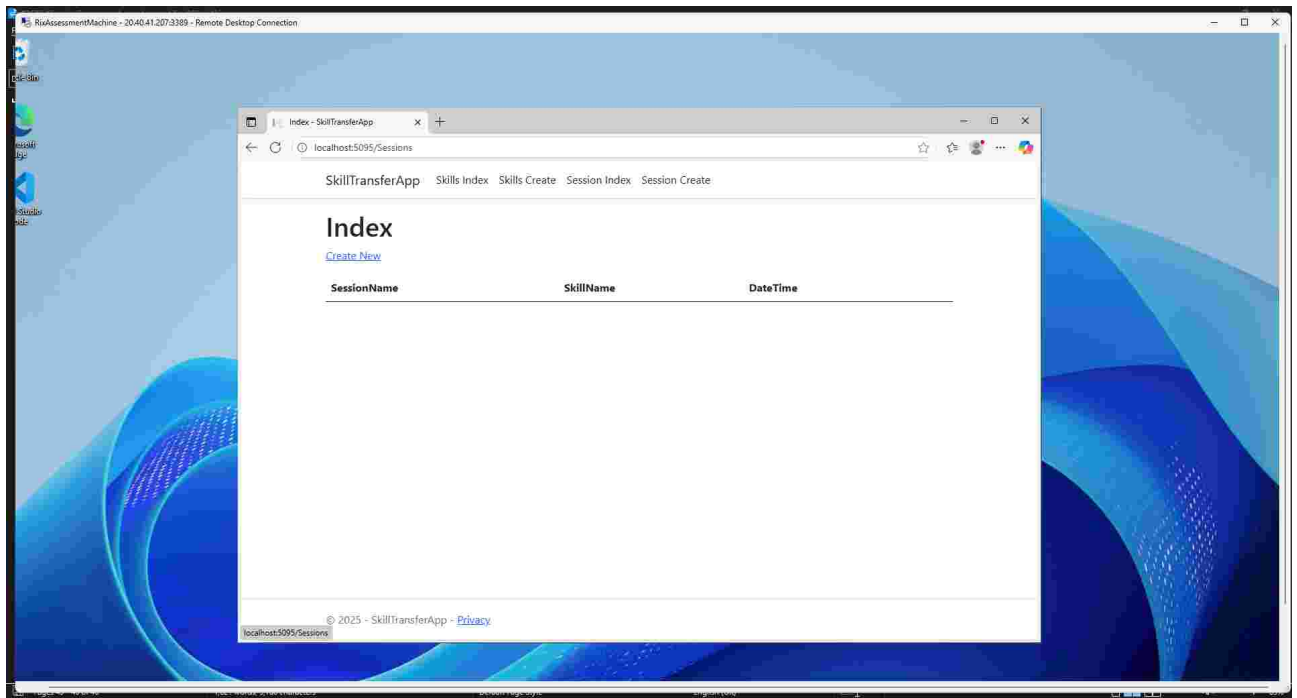
Extract the project and run it



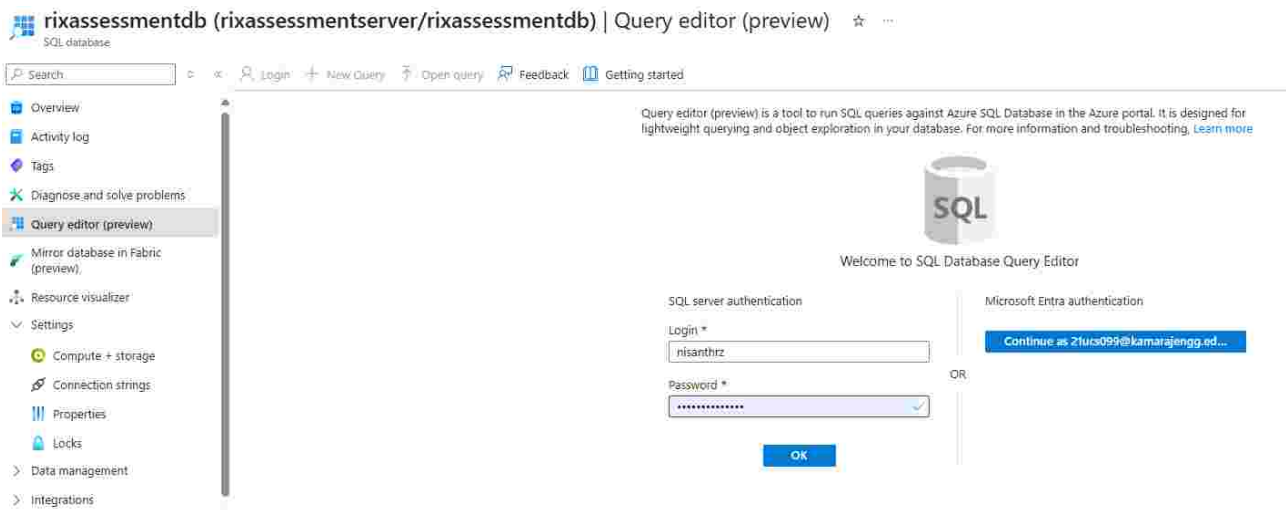


Output inside the VM





CRUD Operations and SQL Reflections

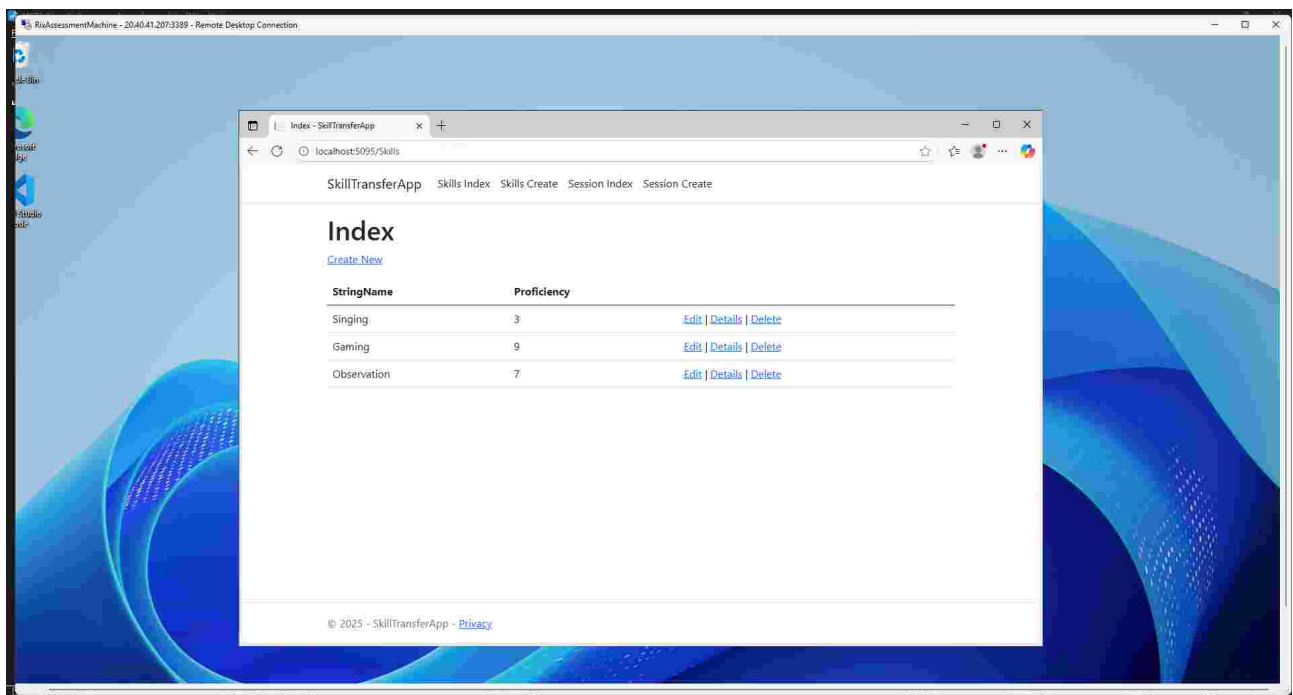
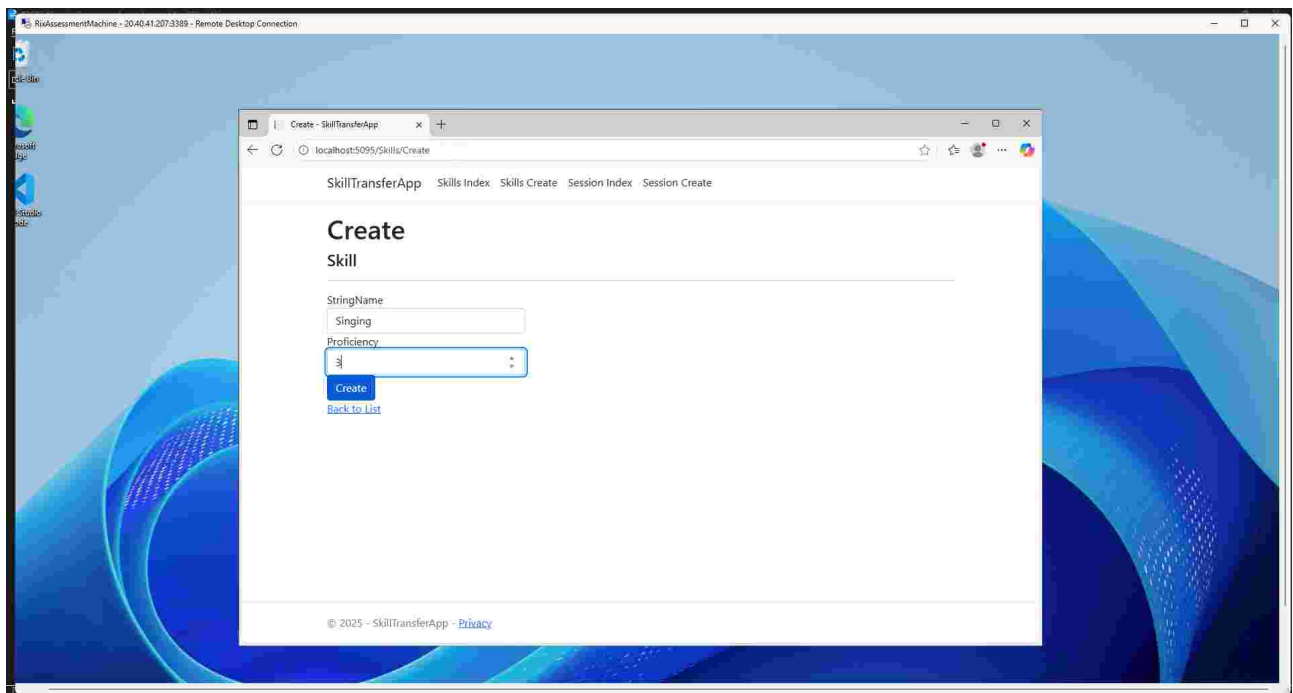


Skills

Initial



Create



Reflection

 Run  Cancel query  Save query  Export data as  Show only Editor

```
1 Select * From [dbo].[Skills];
```

Results Messages

 Search to filter items...

Id	StringName	Proficiency
2	Singing	3
3	Gaming	9
4	Observation	7

Edit

Initial

StringName	Proficiency	
Singing	3	Edit Details Delete
Gaming	9	Edit Details Delete
Observation	7	Edit Details Delete

Edit

Skill

StringName

Gaming

Proficiency

10

Save

[Back to List](#)

Reflection

Index

[Create New](#)

StringName	Proficiency	
Singing	3	Edit Details Delete
Gaming	10	Edit Details Delete
Observation	7	Edit Details Delete

 Run  Cancel query  Save query  Export data as  Show only Editor

```
1 Select * From [dbo].[Skills];
```

Results Messages

 Search to filter items...		
Id	StringName	Proficiency
2	Singing	3
3	Gaming	10
4	Observation	7

Delete

Initial

Index

[Create New](#)

StringName	Proficiency	
Singing	3	Edit Details Delete
Gaming	10	Edit Details Delete
Observation	7	Edit Details Delete

Delete

Are you sure you want to delete this?
Skill

StringName Singing

Proficiency 3

[Delete](#) | [Back to List](#)

Reflection

Index

[Create New](#)

StringName	Proficiency	
Gaming	10	Edit Details Delete
Observation	7	Edit Details Delete

```
1: Select * From [dbo].[Skills];
```

Results Messages

 Search to filter items...

id	StringName	Proficiency
3	Gaming	10
4	Observation	7

Sessions

Initial

 Run  Cancel query  Save query  Export data

```
1  Select * From [dbo].[Sessions];
```

Results

Messages

Query succeeded: Affected rows: 0

Index

[Create New](#)

SessionName

SkillName

DateTime

Create

Create

Session

SessionName

Take me to a game.

SkillName

Gaming

DateTime

09/26/2025 02:21 PM

Create

[Back to List](#)

Index

[Create New](#)

SessionName	SkillName	DateTime	
Take me to a game.	Gaming	9/26/2025 2:21:00 PM	Edit Details Delete
See the sky	Observation	9/19/2025 12:22:00 AM	Edit Details Delete

Reflection

Run

Cancel query

Save query

Export data as

Show only Editor

1

Select * From [dbo].[Sessions];

Results

Messages

Search to filter items

Id	SessionName	SkillName	DateTime
2	Take me to a game	Gaming	2025-09-26T14:21:00.0000000
3	See the sky	Observation	2025-09-19T00:22:00.0000000

Edit

Intial

Index

[Create New](#)

SessionName	SkillName	DateTime	
Take me to a game.	Gaming	9/26/2025 2:21:00 PM	Edit Details Delete
See the sky	Observation	9/19/2025 12:22:00 AM	Edit Details Delete

Edit Session

SessionName

Co-op Gamin

SkillName

Gaming

DateTime

09/26/2025 02:21 PM



Save

[Back to List](#)

Reflection

Index

[Create New](#)

SessionName	SkillName	DateTime	
Co-op Gamin'	Gaming	9/26/2025 2:21:00 PM	Edit Details Delete
See the sky	Observation	9/19/2025 12:22:00 AM	Edit Details Delete

1 `Select * From [dbo].[Sessions];`

Results Messages

Search to filter items...			
Id	SessionName	SkillName	DateTime
2	Co-op Gamin'	Gaming	2025-09-26T14:21:00.0000000
3	See the sky	Observation	2025-09-19T00:22:00.0000000

Delete

Initial

Index

[Create New](#)

SessionName	SkillName	DateTime	
Co-op Gamin'	Gaming	9/26/2025 2:21:00 PM	Edit Details Delete
See the sky	Observation	9/19/2025 12:22:00 AM	Edit Details Delete

Delete

Are you sure you want to delete this?
Session

SessionName	See the sky
SkillName	Observation
DateTime	9/19/2025 12:22:00 AM

Delete | [Back to List](#)

Reflection

Index

[Create New](#)

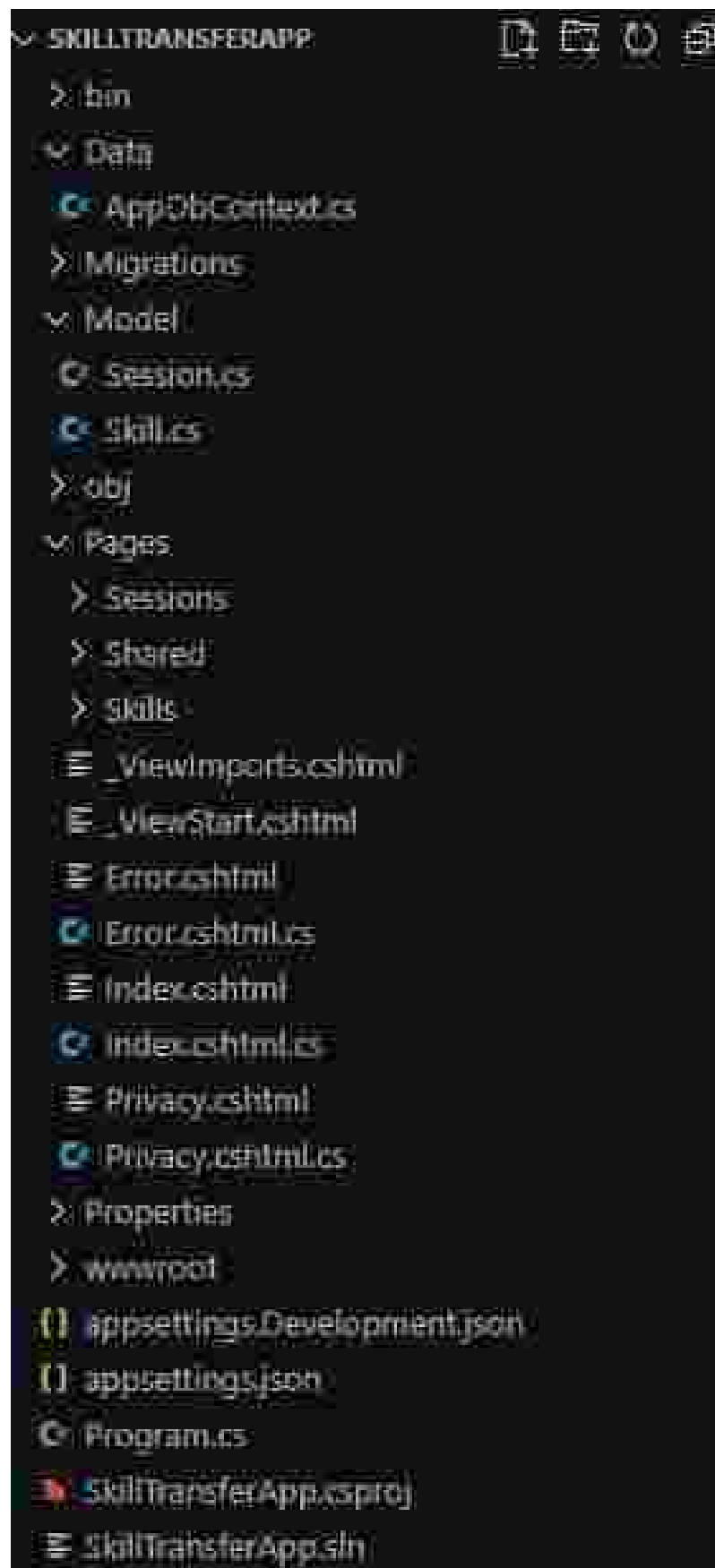
SessionName	SkillName	DateTime	
Co-op Gamin'	Gaming	9/26/2025 2:21:00 PM	Edit Details Delete

```
1  Select * From [dbo].[Sessions];
```

Results Messages

Search to filter items...			
Id	SessionName	SkillName	DateTime
2	Co-op Gamin'	Gaming	2025-09-26T14:21:00.0000000

Code Base



Models

Session

```
using System.ComponentModel.DataAnnotations;

namespace SkillTransferApp.Models
{
    public class Session
    {
        [Key]
        public int Id { get; set; }

        public string? SessionName { get; set; }

        public string? SkillName { get; set; }

        public ICollection<Skill>? Skills { get; set; }

        public DateTime DateTime { get; set; }
    }
}
```

Skill

```
using System.ComponentModel.DataAnnotations;

namespace SkillTransferApp.Models
{
    public class Skill
    {
        [Key]
        public int Id { get; set; }

        public string? StringName { get; set; }

        public int Proficiency { get; set; }

        public ICollection<Session>? Sessions { get; set; }
    }
}
```

AppDbContext.cs

```
using Microsoft.EntityFrameworkCore;
using SkillTransferApp.Models;

public class AppDbContext : DbContext
{
    public AppDbContext(DbContextOptions<AppDbContext> options) : base(options) { }

    public DbSet<Session> Sessions { get; set; }
    public DbSet<Skill> Skills { get; set; }
}
```

Appsettings.json

```
{
  "Logging": {
    "LogLevel": {
      "Default": "Information",
      "Microsoft.AspNetCore": "Warning"
    }
  },
  "ConnectionStrings": {
    "DefaultConnection":
"Server=tcp:rixassessmentserver.database.windows.net,1433;Initial
Catalog=rixassessmentdb;Persist Security Info=False;User
ID=nisanthrz;Password=Password@12345;MultipleActiveResultSets=False;Encrypt=True;Tr
ustServerCertificate=False;Connection Timeout=30;"
  },
  "AllowedHosts": "*"
}
```


Program.cs

```
using Microsoft.EntityFrameworkCore;

var builder = WebApplication.CreateBuilder(args);

// Add services to the container.
builder.Services.AddRazorPages();

// Use SQL Server instead of MySQL
builder.Services.AddDbContext<AppDbContext>(options =>

options.UseSqlServer(builder.Configuration.GetConnectionString("DefaultConnection")
));

var app = builder.Build();

// Configure the HTTP request pipeline.
if (!app.Environment.IsDevelopment())
{
    app.UseExceptionHandler("/Error");
    // The default HSTS value is 30 days. You may want to change this for production
    scenarios, see https://aka.ms/aspnetcore-hsts.
    app.UseHsts();
}

app.UseHttpsRedirection();
app.UseStaticFiles();

app.UseRouting();

app.UseAuthorization();

app.MapRazorPages();

app.Run();
```